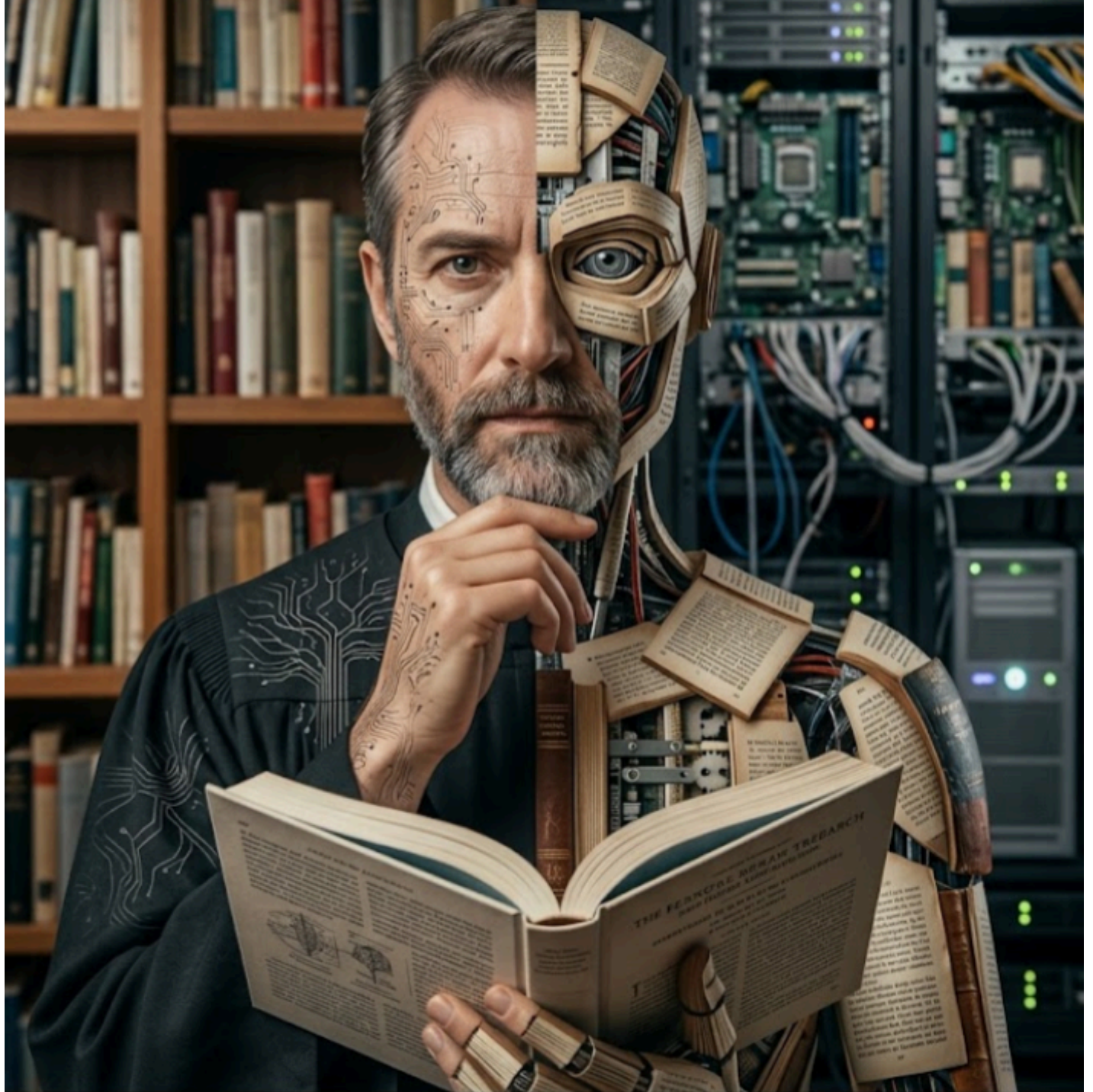


THE LIVING RESEARCH BOOK

A Foresight Monograph on Scientific Work,
Publishing, and Evaluation in 2040



“The future of scientific publishing may contain more text, not less,
while requiring researchers to read less of it.”

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Axiologic Research

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaic. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

Publication Note

This monograph presents a concrete and testable foresight position about scientific work and publishing in 2040. It distinguishes established developments from architectural inference and from the authors' explicit wager. The forecast is intended to guide research, tool construction, institutional experiments, and evaluation rather than to claim inevitability.

The discussions of privacy, intellectual property, regulation, and secure computation are architectural and comparative. They are not jurisdiction-specific legal advice. Any implementation requires legal, ethical, security, and domain review appropriate to the data, partners, location, and intended release.

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Abstract

Scientific publishing has long treated the article as the natural unit of scientific communication. This book argues that the article is better understood as a historically successful interface: a compact container adapted to printing, journal distribution, periodic editorial workflows, scarce reviewer attention, and career systems that need countable outputs. The interface has served science well, but it no longer corresponds to the full object produced by contemporary research. A serious research programme now generates not only claims and prose, but also datasets, source code, computational environments, models, protocols, laboratory records, design decisions, negative results, legal constraints, evolving interpretations, and a web of dependencies on prior work. Most of this structure is either omitted from the article or scattered across repositories that the article can only point toward.

Artificial intelligence creates two apparently opposed futures for this system. In the first, scientific communication becomes radically shorter and more atomic. Researchers publish compact statements of novelty, modular claims, methods, data, and analyses; AI reconstructs the background and explains the result to each reader. This direction is visible in nanopublications, modular publishing platforms, research objects, and machine-actionable metadata. In the second future, which this monograph develops, serious research programmes publish objects that are much larger than articles: living research books that explain prerequisites, reconstruct the literature, document methods and evidence, preserve the evolution of ideas, and expose multiple paths through the work. AI does not make such books burdensome because human readers no longer need to consume them linearly. It can generate a ten-minute briefing for one reader, a rigorous methodological path for another, an executable tutorial for a third, and a reviewer-oriented map of claims and evidence for a fourth.

The central claim is that these futures are not alternatives. The most plausible architecture is atomic underneath and monographic above. Research is represented internally as versioned claims, evidence, code, data, decisions, provenance, contribution records, and policy labels. From this structured substrate, a publication system compiles a coherent, extensive, human-facing research book.

Articles survive, but increasingly as release notes, priority records, or concentrated novelty statements attached to a larger and more persistent scientific object.

This book advances a concrete forecast for 2040. Major research projects and long-lived programmes will commonly maintain AI-assisted living research books, often extending to hundreds or thousands of pages. These books will have continuously updated branches and periodically frozen annual or biennial editions. They will be accompanied by grounded conversational agents, interactive debates among specialised AI systems, generated tutorials, audiovisual explanations, executable experiments, simulations, and reviewer interfaces. The public book will not, however, be a transparent dump of the research workspace. It will be a curated and auditable publication compiled from a more extensive private environment. Personal data, irrelevant conversations, confidential industrial knowledge, pre-filing inventions, security-sensitive information, and raw institutional history may remain protected. Reviewers may receive a controlled evidence capsule that goes beyond the public edition without exposing the entire workspace. The governing principle is epistemic sufficiency rather than total exposure.

The proposal has consequences far beyond document format. It changes team organisation, because contribution can be recorded at the level of claims, experiments, software, review, synthesis, and maintenance rather than compressed into author order. It changes project reporting, because deliverables, periodic reports, public summaries, and funder dashboards can be generated as views of the same maintained research object instead of being rewritten independently. It changes peer review, because reviewers can interrogate a research world, trace assertions to evidence, reproduce selected computations, and inspect disagreements across versions. It changes careers, because a contribution ledger can recognise work that paper counts systematically undervalue, including validation, curation, correction, infrastructure, negative findings, and intellectual integration.

The forecast is deliberately assertive but not presented as destiny. The living research book can fail. It can become a cathedral of fluent emptiness, an instrument of managerial surveillance, a source of privacy leakage, a mechanism for premature loss of intellectual property, or a vehicle for coherent but false scientific worlds. AI may accelerate the current publish-or-perish system rather than

replace it. Journals, universities, funders, and researchers may prefer countable papers because their limitations also make them administratively convenient. The purpose of this monograph is therefore not to announce an inevitable future. It is to articulate a future worth constructing, specify the institutional and technical conditions under which it could work, and identify the tests by which it should be rejected or revised.

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Preface: A Position, Not a Crystal Ball

This is a foresight monograph. Its aim is neither prediction in the narrow statistical sense nor speculative fiction disguised as policy advice. It begins from technologies, institutional tensions, and publishing experiments that already exist in 2026 and asks what coherent scientific institution could emerge if they were assembled around a different unit of communication. The unit proposed here is the living research book: a versioned, AI-maintained, selectively disclosed, multimodal and executable representation of a research project or programme.

The argument emerged from a change of mind. The first intuition was that artificial intelligence would make scientific articles much shorter. A researcher would publish the irreducible delta: the idea, the evidence, the novelty, and perhaps the principal limitations. Everything else would be reconstructed on demand. An AI assistant could explain the prerequisite mathematics, retrieve the relevant literature, translate the contribution for another discipline, and help a reviewer verify positioning. The long introduction and ceremonial literature review would become an inefficient residue of a period in which every paper had to carry enough context for a human reader arriving alone.

That future remains plausible, and parts of it are desirable. Yet it overlooks a second effect of AI. When producing, maintaining, transforming, and navigating text becomes inexpensive, severe compression is no longer required at the source. The source can become richer while the reading experience becomes shorter. A project can preserve a detailed account of its concepts, methods, assumptions, rejected paths, evidence, and dependencies, while each person receives only the path appropriate to a particular question. In other words, artificial intelligence can decouple the size of the scientific record from the amount any individual must read.

Once this decoupling is taken seriously, the article loses its privileged position. A paper may still be useful for announcing a contribution, establishing priority, passing a certification threshold, or entering a conventional citation network. But it is no longer the best container for the scientific object itself. The more ambitious alternative is not simply a long paper. It is a maintained research world with a book as its canonical human interface.

The word book is chosen deliberately, though not nostalgically. A good monograph carries an obligation that a paper can often evade: it must integrate. It cannot resolve every conceptual gap by placing a row of references at the end of a sentence. It must explain how ideas connect, identify what is assumed, allocate space according to intellectual need, and build a path through the argument. The future object proposed here inherits this obligation without inheriting the fixed, linear, single-author form of the printed volume. It may be thousands of pages long, but it is not expected to be read from page one to page one thousand. It may produce a printed or PDF edition, but its deeper structure is computational, versioned, and conversational.

The forecast is intentionally concrete. By 2040, a serious multi-year research programme should be able to maintain a living book that explains its domain, positions its claims, links every important assertion to evidence, records the status of replication, and generates specialised interfaces for readers, reviewers, funders, collaborators, students, regulators, and industrial partners. The system should generate frozen annual or biennial editions that remain citable even as the living branch continues to change. It should support interactive tutorials, videos, simulations, code execution, and discussions among AI agents with explicit roles. It should make the programme easier to understand than a folder containing two hundred papers, not merely larger.

At the same time, the forecast rejects a naive equation between scientific openness and exposure of the entire workspace. A laboratory or project environment contains material that a reviewer does not need and the public should not receive: personal data, confidential correspondence, security details, undocumented suspicions about collaborators, commercially valuable know-how, patentable inventions before filing, licensed datasets, and failed explorations whose disclosure creates risk without adding epistemic value. Scientific accountability requires enough access to evaluate the claim, not universal visibility into every act that preceded it. The living book must therefore be generated through controlled selection, redaction, abstraction, and provenance-preserving compilation.

This distinction is not a concession added after the vision. It is part of the design. The private research workspace, the controlled review capsule, and the public living book are different objects with different access policies. Their relationship must be

explicit and auditable. The system should be able to prove that a public figure was derived from a particular analysis without publishing the underlying personal data. It should be able to let a confidential reviewer inspect protected evidence without allowing that evidence to train an external model. It should be able to establish the timestamp and integrity of an invention while delaying public disclosure until an IP decision has been made. The future of open science is not indiscriminate openness; it is governed accessibility with persistent provenance.

The book also takes a position on scientific careers. Publish or perish is often criticised as if it were merely a stressful slogan or an imperfect metric. The deeper problem is architectural. When the countable unit of evaluation is the paper, research is pushed toward paper-shaped fragments. Teams compete for first authorship, reusable infrastructure is undervalued, negative results disappear, and integration becomes a cost borne by readers. Artificial intelligence can make this worse by increasing the rate at which paper-shaped outputs are produced. A 2026 study of more than forty million papers reported a troubling association: AI adoption correlated with increased productivity and impact for individual scientists while the collective field of inquiry became narrower (Hao et al., 2026). The finding does not prove that AI inevitably creates monoculture, but it demonstrates that local acceleration can coexist with global intellectual contraction.

The alternative developed here is not evaluation without metrics. It is evaluation around richer objects. Contribution records can show who formulated a hypothesis, designed a protocol, built software, curated data, performed an adversarial check, discovered a contradiction, corrected an error, maintained a benchmark, or wrote the integrating explanation. Frozen editions can provide stable assessment points without pretending that a research programme stops evolving. Qualitative review can be supported by quantitative indicators that are tied to the work itself rather than inherited from the prestige of the venue.

Some readers will find the forecast too ambitious. Others will think it insufficiently radical because it retains books, editions, peer review, institutions, and human responsibility. The middle position is deliberate. The proposal assumes that the scientific record will become more machine-actionable and that AI systems will perform much of the maintenance, synthesis, checking, and interface generation. It also assumes that research remains a social institution in which responsibility,

priority, secrecy, disagreement, judgment, and trust cannot simply be delegated to a model. The living research book is therefore neither a conventional monograph with a chatbot attached nor an autonomous database that no one is accountable for. It is an institutional object designed for human and machine collaboration.

The reality of 2040 may contradict this forecast. That is not an embarrassment. Serious foresight should make claims specific enough to fail. It should guide tool building, research programmes, and institutional experiments in the present, while leaving a trail of assumptions that future evidence can challenge. We are not neutral observers of a future that will happen independently of us. By choosing what to prototype, fund, standardise, and reward, we participate in constructing it.

Argument in Brief

The paper is not science itself. It is a compression and coordination device that became institutionally dominant because it fitted the technologies and incentives of its era. Contemporary research exceeds it in both complexity and duration. The gap is visible whenever an article points to data in one repository, code in another, a protocol elsewhere, a preprint with a different version history, supplementary files that cannot be searched coherently, and tacit knowledge that remains in the heads of people who may soon leave the project.

Digital publishing has already produced partial alternatives. Research objects package data, software, workflows, and metadata. FAIR principles require that research artefacts become findable, accessible under appropriate authorisation, interoperable, and reusable. RO-Crate provides a lightweight way to package files with machine-readable contextual metadata (Wilkinson et al., 2016; Research Object Community, 2025; Soiland-Reyes et al., 2022). Living systematic reviews update syntheses as evidence changes. Executable books connect narrative and computation. Nanopublications make claims and provenance citable at a granular level. Modular platforms such as Octopus and ResearchEquals detach stages of research from the single paper. Living-book experiments challenge the assumption that a scholarly book must be fixed, linear, and authored once.

These movements are usually treated as distinct innovations. The thesis of this monograph is that AI can make them components of a single publication architecture. The underlying research workspace can represent granular objects and relationships. A publication compiler can transform those objects into a coherent book and a family of derivative interfaces. The book is living because it is maintained as the research evolves, but it also produces frozen editions because scholarship needs stable citation, accountability, and historical comparison.

The decisive economic change is that source abundance and reader economy no longer conflict. A programme may publish a thousand-page edition because the reader does not need to read a thousand pages. AI can answer a narrow question using the relevant source passages, build a prerequisite path, compare an old and new edition, generate a tutorial, or stage a structured disagreement between

specialised agents. A long source is valuable when it improves grounding, not when length is treated as a proxy for importance.

By 2040, the most mature form of this system will be a research environment rather than a file. Its public face will include a canonical living book, claim-evidence graphs, executable demonstrations, contribution records, review histories, simulations, audiovisual explanations, and grounded conversational agents. Its private side will contain a richer and less polished workspace. Between them will be controlled evidence vaults and review capsules. The architecture will support selective disclosure: reviewers can receive what is necessary to test claims without gaining unrestricted access to personal data, trade secrets, licensed datasets, or the full history of internal deliberation.

The transformation changes scientific work. Writing no longer begins after research; structured explanation and provenance are maintained throughout it. All agents monitor the literature, test internal consistency, compare claims with evidence, detect missing prerequisites, generate candidate explanations, preserve negative results, and prepare review views. Human researchers remain responsible for deciding what questions matter, what evidence is persuasive, what risks are acceptable, and what should be made public.

The transformation changes collaboration. Instead of assigning most credit through author order, the system can record contributions continuously and at useful granularity. This makes integrative work, validation, software maintenance, data stewardship, and correction more visible. It may reduce some forms of competition, but it can also create intrusive surveillance. A contribution ledger must therefore record scholarly responsibility without becoming a productivity panopticon.

The transformation changes project reporting. Funders currently ask for deliverables, periodic reports, milestone updates, publication lists, data-management records, exploitation plans, and public summaries, often in separate templates. In the proposed architecture these are generated views of the same maintained object. Reporting becomes a query over evidence and commitments rather than a ritual of duplicative writing. The funder can inspect which objectives changed, which claims remain unsupported, what risks materialised, and how the programme responded.

The transformation changes peer review. Reviewers should not be expected to read an entire thousand-page book, nor should they receive the full laboratory workspace. They should receive an interface that exposes the contribution map, literature positioning, claim-evidence relationships, reproducibility status, unresolved contradictions, version changes, and selected protected evidence. AI performs search, checking, and navigation; humans perform judgment. Review can become continuous and claim-specific rather than a single opaque verdict on a static manuscript.

The transformation changes career evaluation. Paper counts and journal prestige reward fragmentation because the unit of assessment determines the unit of production. A living research book makes it possible to assess durable contributions to a programme: conceptual advances, tools, datasets, experiments, syntheses, reviews, corrections, mentoring, and maintenance. DORA, the Leiden Manifesto, CoARA, and CRedit already supply parts of the normative and descriptive vocabulary for such a transition (DORA, 2012; Hicks et al., 2015; CoARA, 2022; CRedit, 2026). What is missing is an object rich enough to carry those reforms operationally.

The vision can fail in predictable ways. AI can generate enormous documents that disguise trivial contributions. A coherent book can make a false theory appear more credible. Selective disclosure can become selective deception. Protected review environments can still leak. Automated synthesis can narrow inquiry by repeatedly retrieving the same visible literature. Proprietary platforms can own the interface to a field. Fine-grained contribution records can encourage local gaming and institutional control. Annual editions can become bureaucratic deliverables rather than living knowledge.

For these reasons, the living research book requires governance. Major claims need explicit status. Derivations need provenance. Editions need immutable identifiers. Corrections must remain visible. AI-generated material must be attributable to models, prompts, sources, and human approvals. Access decisions need policy and audit. IP review must occur before public release. Confidential reviewers need secure tools that do not reuse protected content for training. Libraries and public infrastructures must preserve both human-readable editions and machine-actionable structures.

The forecast is therefore conditional but firm. The article will not vanish by 2040, but for serious, long-lived programmes it will cease to be the primary home of scientific understanding. It will become a compact event in the history of a larger object. The book, rebuilt as a living, executable, conversational, selectively disclosed research world, will become the more adequate unit.

Table 1. The institutional shift proposed by this monograph.

Dimension	Implication for the living research book
Primary unit	From the bounded article to the persistent research programme and its versioned living book.
Temporal model	From publication as a terminal event to a live branch with frozen annual or biennial editions.
Internal structure	From prose with attachments to claims, evidence, data, software, decisions, provenance, contributions, and policies.
Reader experience	From compulsory linear reading to adaptive routes, grounded questions, tutorials, simulations, and audiovisual explanations.
Peer review	From reconstructing context manually to inspecting claim-evidence maps and controlled review capsules.
Transparency	From either sparse disclosure or indiscriminate repository dumps to selective epistemic disclosure.
Credit	From author order and paper counts to qualitative portfolios backed by contribution records.
Project reporting	From repeatedly rewritten documents to authorised projections of maintained knowledge.

CHAPTER 1

1. THE PAPER WAS AN INTERFACE, NOT SCIENCE

1.1 The historical success of a constrained form

Scientific articles are so deeply embedded in the organisation of research that they are easily mistaken for natural units of knowledge. A result becomes real when it is a paper. A project is productive when it yields papers. A researcher is visible through a list of papers. A field is mapped through citation relations among papers. Even criticisms of publishing often preserve this ontology: they ask for faster papers, more open papers, better reviewed papers, or papers with reusable data. The assumption that science must be represented as a sequence of bounded textual objects is rarely examined.

That assumption is historically understandable. The scientific paper solved several difficult coordination problems at once. It established a compact record that could be copied, transported, catalogued, reviewed, and cited. It made priority disputes more manageable by attaching dates, authors, and venues to claims. It allowed journals to assemble heterogeneous contributions into periodic issues and libraries to preserve a recognisable record. It also created a practical contract between author and reader. The author was not expected to reproduce the whole field; the reader was expected to arrive with substantial background knowledge and to follow references when more context was required.

The form acquired conventions that increased its efficiency. The title promised a bounded contribution. The abstract offered a compressed preview. The introduction positioned the problem. Methods described what had been done, results stated what had been observed, and discussion connected the observations to larger claims. Disciplinary variations were large, but the central logic remained stable: reduce an extended and often chaotic inquiry to a portable argumentative package.

This compression was not merely aesthetic. Printed pages, journal space, postal distribution, editorial labour, and reader attention were costly. The paper made it possible to move an intellectual object through institutions that could not economically distribute an entire laboratory archive. Even after printing ceased to be the main cost, the inherited form continued to solve other problems. Editors could request two or three anonymous reports on a document of manageable size. Promotion committees could count outputs. Citation databases could index them. Grant agencies could ask for a publication list. The article was not only a medium; it became a unit of administration.

Its success should therefore be treated with respect. Many proposals for post-paper science fail because they underestimate the number of institutional functions compressed into the familiar format. Replacing a PDF is easy. Replacing priority, certification, discoverability, accountability, career credit, archiving, and shared reading conventions is difficult. The persistence of papers is not evidence that researchers are technologically conservative. It reflects the stability of a coordination equilibrium.

Yet an equilibrium can become maladaptive when the object being coordinated changes. Contemporary research increasingly involves distributed teams, long computational pipelines, large datasets, external services, specialised software, evolving regulatory constraints, and results that depend on thousands of decisions not visible in the final prose. The paper remains compact because its institutional functions still reward compactness, not because the underlying work is compact.

The central distinction of this book begins here. Science is an activity and an evolving structure of claims, methods, evidence, instruments, disagreements, and communities. The paper is one interface to that structure. An interface can be excellent and still cease to be adequate. Treating it as contingent rather than natural makes it possible to ask what interface would be designed for a world in which machines can read continuously, maintain provenance, generate explanations, and adapt presentation to the reader.

The question is not whether the paper has failed. The question is whether its original virtues have become constraints. Portability once required omission. Reviewability once required a small object. Citation once required a bounded work. Career evaluation once required administratively countable units. Digital and

AI-supported environments weaken each of these requirements. A research object can be large yet navigable, continuously changing yet periodically frozen, granular yet coherently explained, private in some layers and public in others, machine-actionable without becoming unreadable to people.

The paper also shaped expectations about scientific voice. It rewards a retrospective narrative in which the problem appears well defined, the method appears selected for clear reasons, and the result appears as the orderly consequence of a plan. Actual research is more recursive. Questions change after methods fail. Concepts are redefined. Data reveal artefacts. Literature is rediscovered late. A useful negative result alters the programme but never becomes a publishable story. The paper's clarity is often achieved by deleting the path that made understanding possible.

This deletion is not necessarily dishonest. Communication requires selection. No reader wants a chronological transcript of every meeting or failed command. The problem is that the paper offers only one level of selection and makes its omissions difficult to inspect. It cannot distinguish between irrelevant mess and relevant history that would change interpretation. The proposed living research book will preserve this distinction by separating the private workspace from the curated scientific record. It does not demand total disclosure; it demands that important dependencies and decisions have somewhere to live.

The article's boundedness also encourages a misleading model of completion. A paper is submitted, revised, accepted, and published. It may later be corrected or retracted, but the normal expectation is finality. Research programmes are not final in this way. Their concepts, datasets, code, and interpretations continue to change. New literature can weaken the novelty claim. A replication can alter confidence. An industrial deployment can reveal assumptions that were invisible in the original experiment. A static paper remains historically valuable, but it becomes an increasingly incomplete interface to a living body of work.

By 2040, the paper will still exist because some scientific events benefit from compact certification. A new theorem, measurement, benchmark result, or conceptual distinction may deserve a concise release. The prediction advanced here is narrower and stronger: for serious research programmes, the paper will no

longer be the main object through which the work is understood. It will point into, and be generated from, a larger maintained system.

1.2 The mismatch between research work and publishable narrative

The mismatch can be seen by comparing what a project actually produces with what appears in its publications. A computational project may create data-ingestion scripts, model configurations, training runs, failed architectures, evaluation suites, issue trackers, security decisions, licence analyses, notebooks, deployment experiments, and conversations that clarify requirements. A laboratory project may create samples, instrument configurations, calibration records, laboratory protocols, images, quality-control decisions, negative measurements, and tacit procedures learned through repetition. A social-science project may create consent procedures, interview guides, coding frameworks, field notes, contested interpretations, and records of contextual decisions. The article compresses these heterogeneous materials into a linear argument.

The materials do not disappear. They become distributed. Some enter supplementary information. Some are deposited in data or software repositories. Some remain in institutional storage. Some are preserved only because a diligent doctoral researcher keeps a private directory. Some are lost when a contract ends. The resulting publication ecosystem is not truly paper-centred; it is paper-fronted and archive-fragmented.

Research objects and research compendia emerged as responses to this fragmentation. FAIR principles explicitly address the needs of both humans and machines to discover, access, integrate, and analyse research data together with associated algorithms and workflows (Wilkinson et al., 2016). The phrase "accessible" in FAIR is often misunderstood as universally open; the principles allow access under appropriate authorisation. RO-Crate provides a lightweight way to package files with contextual metadata in human- and machine-readable forms (Research Object Community, 2025; Soiland-Reyes et al., 2022). Executable research compendia connect narrative, code, data, and computational environments so that at least some results can be reproduced rather than merely described.

These developments are important, but they often remain adjuncts to the article. The paper tells the story, while the repository contains the machinery. The user must reconstruct the relationship between them. A figure may have been generated by a script, but the paper does not necessarily expose which version, which parameters, which input snapshot, or which manual transformation intervened. Provenance is present in pieces but not designed as a first-class relation.

The problem becomes more severe as AI systems enter research. An agent may generate hypotheses, retrieve literature, write code, execute experiments, analyse outputs, revise text, and criticise conclusions. By 2026, end-to-end automation has already been demonstrated in restricted domains of machine-learning research, and multi-agent systems are being tested in scientific discovery (Lu et al., 2026). These systems do not eliminate human work, but they multiply the number of intermediate artefacts. A single result may depend on a chain of model outputs, tool calls, retrievals, simulations, and human approvals. A prose methods section cannot carry enough information to make that chain understandable.

This creates a paradox. AI makes research production more scalable, but the paper remains a narrow output channel. The temptation is to use AI to squeeze more work through the same channel: more papers, faster literature reviews, more polished prose, more submissions. The 2026 Nature study by Hao and colleagues is relevant precisely because it reports an association between AI use and increased individual scientific productivity and impact, while also finding contraction in collective topical diversity and follow-on engagement (Hao et al., 2026). The causal mechanisms remain open to debate, but the institutional warning is clear. Accelerating paper production can improve local performance while narrowing the global exploration of science.

The mismatch is not only technical. It affects epistemic quality. A paper's literature review must be selective, but selection can hide contested assumptions. Its methods must be concise, but concision can omit practical conditions required for replication. Its discussion must tell a coherent story, but coherence can suppress anomalous evidence. Its authorship line must be short, but the work may depend on dozens of forms of expertise. Its finality can hide uncertainty that continues to evolve.

The traditional response is to strengthen reporting guidelines. This is valuable. Domain-specific checklists can improve transparency and comparability. Yet every additional checklist item competes for space and author attention. The paper becomes longer without becoming structurally richer. A living research book approaches the problem differently. It does not require every reader-facing view to contain everything. It requires the underlying scientific object to contain enough structured information that different views can be generated reliably.

Consider a claim that a new model improves diagnostic accuracy. The public abstract may state the result and principal limitations. A clinical reader may need cohort definition, performance by subgroup, calibration, and comparison with standard care. A machine-learning reviewer may need splits, preprocessing, hyperparameters, uncertainty estimates, and baseline implementation details. A privacy reviewer may need information about consent, legal basis, data minimisation, and re-identification risk. A patent attorney may need to know which technical elements were publicly disclosed and when. A programme evaluator may need to know whether the result fulfils a contractual objective. No single article can optimise for all these readers without becoming unreadable.

The living research book does not solve this by placing every detail in one uninterrupted narrative. It treats the book as a canonical source with multiple reading paths. The same claim can be linked to distinct evidence and explanation layers. AI assists navigation, but the source remains inspectable. A reader can request a concise answer and then descend into supporting material. The system can reveal uncertainty, alternative interpretations, and version history rather than generating a smooth response from an opaque corpus.

This architecture also changes the status of negative results. In a paper-centred system, a failed experiment is often valuable only if it can be reframed as a publishable finding. In a programme-centred system, it can change the evidence graph, close a path, inform design, and prevent repetition. The result may not deserve a public article, but it deserves a place in the maintained knowledge object. Some negative results can remain internal; others can be published when their external value exceeds confidentiality or IP concerns. The important point is that the system can represent them without requiring an artificial narrative.

Tacit knowledge remains a hard boundary. Not everything a skilled researcher knows can be captured in prose or metadata. A protocol may work because of sensory judgment, material familiarity, or domain intuition that is difficult to formalise. The 2026 perspective by Rainford and colleagues on knowledge preservation in big science and AI emphasises the risk that practical know-how and historical versions disappear as projects and personnel change (Rainford et al., 2026). A living research book cannot fully encode expertise, but it can create incentives and tools for preserving more of it through demonstrations, annotated procedures, decision rationales, and interactive tutorials.

The goal is not archival maximalism. Recording everything is neither possible nor desirable. The goal is to make the scientific record correspond more closely to the structure of the work. This requires decisions about relevance, privacy, cost, and future reuse. AI can lower the cost of curation, but it cannot eliminate the need for governance. A project must define which events become durable claims, which artefacts become evidence, which discussions matter to interpretation, and which material remains outside the publication boundary.

The difference from a repository is therefore conceptual. A repository stores things. A living research book explains why they matter, how they relate, what they support, and what remains uncertain. It is not a warehouse attached to a paper; it is the maintained intellectual account of the programme.

1.3 When machines become primary readers

The design of scientific communication changes when machines become routine readers. This does not mean that AI systems replace scientists as the ultimate audience. It means that retrieval, filtering, comparison, summarisation, translation, and preliminary checking increasingly occur through computational intermediaries. A researcher may encounter a result first as an answer generated from literature rather than as a title in a table of contents. A reviewer may ask an assistant to compare a manuscript with prior methods. A funder may query a portfolio for evidence of progress. A student may learn a concept through an adaptive explanation derived from several sources.

The emergence of systems such as OpenScholar illustrates the shift (Asai et al., 2026). OpenScholar retrieves from a large corpus of open-access papers and

synthesises citation-backed answers. Its significance is not that it has solved literature synthesis. It has not. Citation validity, coverage, source selection, and the interpretation of disagreement remain difficult. Its significance is that the literature is beginning to function as a queryable substrate rather than a sequence of documents that each person must locate and read manually.

Once this interaction becomes normal, documents are evaluated partly by how well machines can interpret them. Machine-actionable metadata, persistent identifiers, provenance, explicit licences, structured references, and stable versions cease to be secondary infrastructure. They become part of scientific rhetoric because they determine what can be found, compared, and verified.

A paper designed only for human reading is a poor substrate for this environment. Claims may be implicit. Tables may be images. Methods may use inconsistent terminology. Supplementary files may lack identifiers. Corrections may not propagate to downstream systems. The document can remain elegantly readable while being computationally ambiguous. Conversely, a database optimised only for machines can be semantically brittle and unusable for human reasoning. The living research book must hold both requirements together.

The core design principle is dual addressability. Every important component should be available as part of a coherent human argument and as a machine-addressable object. A claim should have a stable identifier, a textual formulation, a status, provenance, supporting and opposing evidence, scope conditions, and relationships to other claims. An experiment should have narrative context and structured parameters. A contribution should be explained and represented in a standard role vocabulary. A chapter should be readable as prose and decomposable into traceable source units.

This dual structure enables what may be called adaptive scholarly resolution. The same programme can be represented at different levels without losing the path between them. A one-paragraph summary points to a ten-page explanation. The explanation points to claims. Claims point to evidence. Evidence points to data, code, protocols, and review decisions. The reader can stop at the level appropriate to the task. The AI system can compress, but its compression remains accountable to a stable source.

The economic consequence is counterintuitive. Scientific publishing may contain more text while demanding less reading. Source completeness and consumption efficiency become compatible. The long book is no longer justified because every reader should traverse it. It is justified because a well-grounded interface needs a rich source from which to construct reliable paths.

This also reverses a common fear about AI-generated text. The concern is often that models will flood science with verbose, repetitive prose. That risk is real. The answer, however, is not necessarily to impose shorter source documents. It is to distinguish generated surface area from epistemic density. A living book must expose which passages contain original claims, which synthesise established knowledge, which explain prerequisites, which are generated tutorials, and which are interpretations. Its architecture should make a thousand pages less deceptive than twenty, not more.

Machine reading introduces new forms of dependency. If AI assistants mediate access to science, the ranking and retrieval systems behind them can shape what is visible. The most frequently cited or best indexed sources may be repeatedly selected, reinforcing intellectual concentration. Proprietary models may answer from inaccessible corpora, making verification difficult. Generated summaries may smooth disagreement into a false consensus. Research communities could become dependent on a few platforms that control the interface to their own knowledge.

The living research book offers only a partial response. By maintaining explicit sources, versioned editions, and open machine-readable structures, it can reduce dependence on opaque retrieval. Institutions can run local or domain-specific assistants grounded in the canonical edition. Libraries can preserve the source and provide independent interfaces. Competing agents can produce different interpretations over the same evidence graph. None of this guarantees pluralism, but it makes the mediation layer more inspectable.

The role of the human reader also changes. Reading becomes less about locating every relevant sentence and more about choosing questions, evaluating explanations, identifying missing perspectives, and deciding when to inspect primary evidence. This is not a reduction of scholarly skill. It is a redistribution. The researcher who accepts every generated synthesis uncritically becomes vulnerable to confident error. The researcher who refuses all computational mediation

becomes unable to navigate the scale of the record. Scientific literacy in 2040 will include the ability to interrogate machine-mediated knowledge.

Peer review offers a concrete example. Current policies from major publishers correctly warn reviewers not to upload confidential manuscripts into unsecured public AI tools (Nature Portfolio, 2026). The prohibition reflects privacy, confidentiality, and intellectual-property risks. Yet it also reveals the direction of travel. Reviewers already want computational support, and publishers are considering safe systems. A future review environment can provide a bounded AI assistant that operates over the submitted book and authorised evidence, keeps protected content within the review boundary, records its operations, and never delegates the final judgment.

The machine as primary reader is therefore not an autonomous scientific authority. It is a pervasive interpretive layer. The quality of science will depend partly on what that layer is allowed to see, how it represents uncertainty, what provenance it preserves, how it is evaluated, and who controls it. A paper-centred system leaves these questions to external platforms. A living research book treats them as design requirements of publication itself.

The chapter's conclusion is deliberately modest in wording and radical in implication. The paper remains useful, but its adequacy was contingent on a world of scarce distribution and predominantly human navigation. A world of abundant computation and machine-mediated reading needs a different primary object. The rest of this book describes one candidate and the institutional wager it requires.

2. TWO FUTURES OF SCIENTIFIC PUBLISHING

2.1 The attractive logic of the atomic future

The first plausible future of scientific publishing is not the book but the claim. It begins from a reasonable diagnosis: the article combines too many functions and forces too many stages of research into a single narrative. A research problem, hypothesis, method, dataset, analysis, interpretation, and review are intellectually different objects. They can appear at different times, be contributed by different people, and deserve different forms of scrutiny. Why should they wait until a complete paper can be assembled?

Nanopublications provide one answer (Kuhn et al., 2016). A small assertion is published together with provenance and publication information. The unit is machine-actionable and citable. Modular platforms provide another. Octopus describes itself as a primary research record in which researchers can publish work in full detail and receive credit without forcing every contribution into the impact and word-count expectations of a conventional article (Octopus, 2026). ResearchEquals similarly treats research modules as independently publishable units (ResearchEquals, 2026). These systems respond to a real weakness of paper-centred communication: valuable work can be delayed or lost because it does not fit a complete story.

The atomic future has several advantages. It can establish priority early. It can expose methods and data before a final interpretation is available. It can separate criticism of one component from rejection of an entire paper. It can make negative findings and replications first-class outputs. It can support contribution credit at a useful level of granularity. It can make research more composable, because later work can cite a method, dataset, or claim directly rather than referring ambiguously to a whole document.

Artificial intelligence appears to strengthen this logic. If machines can reconstruct context, why repeat it in every publication? A researcher could publish a compact novelty object: the precise difference from prior knowledge, the evidence, the confidence, the scope, and the links required for verification. A reader's assistant could retrieve prerequisites and explain the result at the desired level. The reviewer could ask a system to compare the claim with existing literature. Ceremonial prose would shrink, and the scientific record would become cleaner.

This model also addresses the increasing mismatch between publication speed and research speed. A programme may produce meaningful intermediate results for years before a major article appears. Atomic publication allows the record to follow the work continuously. The idea resembles version control in software: small, attributable changes are integrated into a larger history rather than released only as occasional monoliths.

There is a strong epistemic argument for this granularity. Scientific disagreements often concern one link in a chain. A dataset may be sound while an interpretation is weak. A method may be useful even if a benchmark claim fails. A hypothesis may be novel even if the first test is inconclusive. The paper binds these elements so tightly that a binary accept-or-reject decision obscures partial value. Atomic objects allow more precise evaluation.

The model is also compatible with research objects and FAIR infrastructure. A claim can refer to an identifiable dataset, an executable workflow, a software version, and a protocol. Machines can traverse the relations. Credit can follow the specific objects reused. Literature synthesis can operate over structured claims rather than extracting them imperfectly from prose.

Yet the atomic future has a serious weakness: decomposition does not automatically produce understanding. A field represented as millions of small objects may be more machine-readable and less intelligible. Relations among claims can be explicit, but their significance still requires conceptual integration. A researcher entering the field needs more than a graph of statements. The researcher needs to know why distinctions matter, how vocabulary evolved, which assumptions are contested, what failure modes recur, and how methods fit together.

The paper already delegates much of this work to reviews, textbooks, monographs, and tacit training. An atomic system would increase the need for

synthesis. AI can generate it, but generated synthesis becomes trustworthy only when it is grounded in a sufficiently rich and governed source. If the source consists entirely of claims stripped of interpretive context, the machine must reconstruct the missing conceptual world. The result may be fluent but fragile.

Atomic publication also creates incentives for output inflation. When every method, result, and interpretation can be counted separately, institutions may reward the number of modules rather than their contribution. The unit can become smaller without the metric becoming wiser. Salami slicing does not disappear; it can become formally standardised. Review burden can increase because a stream of small objects still requires certification and relation to the existing record.

Priority becomes another difficulty. A granular claim can establish a timestamp, but scientific priority is rarely a simple race to a sentence. It involves the adequacy of evidence, independence of conception, scope of the contribution, and relation to prior work. Extremely early publication may create noisy priority disputes and reward premature claims. A system designed for rapid atomic release needs maturity states that distinguish conjecture, preliminary evidence, validated result, replication, and superseded interpretation.

There is also a social limit. Science is not only an accumulation of statements. Communities form around programmes, paradigms, instruments, methods, and explanatory ideals. People learn through narratives that organise attention. A perfectly normalised database can record that one proposition supports another, yet fail to convey what it was like to discover the distinction or why a sequence of failures changed the programme's direction. Some of this narrative is unnecessary; some is how understanding is transmitted.

The atomic future should therefore be retained, but not treated as a complete publication philosophy. Granularity is an excellent property of the substrate. It is not necessarily the best property of the interface. The mistake is to assume that because machines benefit from decomposition, human understanding should also be presented as fragments.

2.2 Why artificial intelligence reverses the economy of length

The conventional economics of publishing create a trade-off between completeness and usability. A long document can preserve context but is costly to produce, review, distribute, and read. A short document is easier to circulate but shifts work onto the reader. References are one mechanism for managing the trade-off. Instead of explaining a prerequisite, the author points elsewhere. The reader reconstructs the chain when necessary.

Artificial intelligence weakens both sides of this trade-off. It can assist in producing and maintaining extensive explanations. It can also assist readers in navigating them. The result is not simply cheaper writing. It is a reversal of the relation between source length and reader burden.

Imagine a research programme that has produced a thousand-page book. Under print assumptions, such an object appears excessive. Few reviewers can read it, and most readers will not try. Under an AI-mediated interface, the thousand pages function more like an authoritative domain model. A reader asks for the exact contribution in five paragraphs. Another requests a forty-page tutorial covering prerequisites. A reviewer asks for every claim whose evidence depends on one dataset. A regulator asks for the privacy analysis and data lineage. A collaborator asks how the latest edition differs from the previous one. The source remains large, but consumption is selective.

The principle can be stated precisely: source abundance becomes useful when navigation is cheap and traceable. A large corpus without reliable retrieval is a burden. A large corpus with opaque retrieval is dangerous because the system can choose evidence without accountability. A large, structured, versioned corpus with provenance-aware retrieval can reduce the cost of understanding while increasing the amount available for verification.

This is why the future may contain more scientific text while requiring scientists to read less of it. The additional text is not justified as rhetorical expansion. It is justified as coverage. It explains prerequisites that would otherwise be scattered across references, preserves decisions that affect interpretation, records limitations in detail, and supplies pedagogical paths for different audiences. The AI layer

performs compression at the moment of use rather than forcing authors to compress permanently at publication.

The distinction between permanent and reversible compression is essential. A paper permanently omits much of the context. Later readers cannot recover what was never recorded. An AI-generated summary reversibly compresses a richer source. The reader can descend to details, compare alternative summaries, or inspect the passages that support an answer. Reversible compression is not guaranteed by language models; it must be engineered through citations, stable source units, retrieval traces, and edition control. But it is a plausible design objective.

This reversal also changes the purpose of writing. Researchers have traditionally rewritten similar background sections because each paper must be independently legible. A maintained programme book can explain a concept once, improve the explanation over time, and reuse it across generated views. New results update the relevant sections rather than spawning a nearly duplicate introduction. The cost saved is not only author time. Readers no longer traverse repeated summaries with minor variations.

The same principle applies to literature positioning. A paper usually contains a selective related-work section frozen at submission. A living book can maintain an evolving map of relevant literature, with explicit reasons for inclusion, disagreement, and omission. AI agents can monitor new publications and propose updates. Human researchers approve changes that affect novelty or interpretation. The reader can ask what changed after a particular date.

The danger is obvious: once text becomes cheap, projects may generate vast amounts of low-value prose. A thousand-page book can become a monument to the productivity of its agents rather than to the quality of its science. Length must therefore be treated as a capacity, not an objective. The system should make epistemic density visible. It should distinguish original contributions from established background, verified content from generated explanation, stable conclusions from speculative extensions, and primary evidence from rhetorical synthesis.

A useful design rule is that no generated expansion should obscure the delta. Every edition needs a compact contribution map stating what is new, what changed,

what was corrected, and what remains unresolved. The long book surrounds this map with enough context to make it defensible. The project does not earn credit for pages; it earns credit for contributions that survive scrutiny.

Review also changes under this economy. The reviewer is not asked to read the entire source. The review system constructs a bounded route through it. This route may include the novelty claims, central evidence, methodological details, known limitations, and disagreements detected by agents. The reviewer can expand any component. The book is thus simultaneously more complete than a paper and less demanding as a default review object.

The economics extend to education and transfer. A new team member can receive a generated learning path based on prior knowledge. An industrial partner can receive a version focused on deployment assumptions. A policymaker can receive an explanation of societal implications linked to evidence. A machine can generate a video, interactive demonstration, or simulation from the maintained source. These are not separate publications requiring independent manual maintenance. They are derivative interfaces whose consistency can be checked against the canonical object.

This creates a new publishing value chain. The scarce activity is no longer typesetting or producing text. It is maintaining trustworthy structure. Publishers, libraries, and scientific institutions may compete on certification, preservation, identity, provenance, secure review, model governance, and high-quality interfaces. The living research book is economically plausible only if these trust functions are funded. AI does not make publishing free; it changes what publishing must pay for.

The reversal of length economics is therefore conditional. It depends on structured sources, reliable navigation, explicit provenance, bounded generation, and governance. Without these, more text means more noise. With them, more context can reduce rather than increase the burden on readers.

2.3 Atomic underneath, monographic above

The apparent conflict between atomic publishing and living books dissolves when they are assigned to different architectural layers. Research should be atomic underneath because claims, evidence, data, methods, software, decisions, and contributions need independent identities. It should be monographic above because

people need coherent explanations and programmes need integrated intellectual accounts.

The proposed model can be understood as a compiler. The research workspace produces structured source material. A publication compiler selects authorised components, verifies references, applies privacy and IP policies, resolves version dependencies, and generates a living research book. The book is not manually written from nothing, nor is it an unedited model output. It is a governed rendering of a maintained research object.

At the atomic layer, a claim has a scope, status, provenance, and evidence relation. An experiment has inputs, procedures, outputs, environment, and interpretation. A dataset has lineage, consent constraints, licences, and access policies. A software component has versions, dependencies, tests, and authorship. A review comment has an author or agent identity, a target, a resolution, and a visibility policy. A contribution event records responsibility without necessarily recording every keystroke.

At the monographic layer, these objects become an argument. The book explains why the claim matters, how it arose, what evidence supports it, where it conflicts with previous work, what limitations remain, and what a reader needs to understand before evaluating it. The monographic layer can include prose written by people, prose drafted by agents and approved by people, generated tutorials, diagrams, simulations, and conversations. Its authority comes from traceability to the structured substrate and from accountable editorial decisions.

This architecture makes the article a derivative object. A paper can be generated as a compact release describing a coherent change to the programme. It has its own identifier and frozen text. It points to the relevant edition, claims, evidence, and code. It can be peer reviewed independently when the community requires a familiar unit. What changes is its ontological status: it is no longer the container of the whole result but a certified event in the evolution of a larger research world.

The analogy with software releases is useful but incomplete. A scientific programme is not a codebase, and truth does not follow from successful integration tests. Yet several practices transfer well. Version history preserves change. Branches allow competing interpretations. Issues record unresolved problems. Releases create stable reference points. Dependencies are explicit. Regression

tests detect when a new change breaks an old claim. Scientific equivalents would include re-running benchmark suites, checking whether updated data alter conclusions, and verifying that a rewritten explanation still cites the correct evidence.

The monographic layer adds what software repositories often lack: sustained conceptual narration. It explains the architecture, rationale, and intellectual stakes. A research programme without such integration becomes a collection of artefacts whose relation is known only to insiders. The living book is the institution's obligation to make its knowledge transmissible.

The model also supports plurality. A single programme may maintain more than one legitimate narrative over the same substrate. A technical edition may organise the work by method. A historical edition may trace the evolution of ideas. A teaching edition may order concepts by prerequisite. A policy edition may focus on evidence relevant to decisions. These are not arbitrary summaries if their derivation is explicit and their claims point to the same canonical objects.

Plurality must not dissolve responsibility. A project needs a canonical public account against which it can be evaluated. The canonical living book is not the only view, but it is the version for which the programme accepts editorial and scientific responsibility. Generated views should declare their purpose, source edition, model provenance, and any transformations that could affect meaning.

The relationship between the public book and the private workspace requires special care. Atomic structure does not imply that every atom is public. Some objects contain personal data, confidential know-how, licensed material, security-sensitive details, or internal deliberation that does not affect the published claims. Access policy is therefore part of the substrate. The compiler can produce a public book from public and abstracted objects, a confidential reviewer capsule from a broader authorised set, and an internal edition from the full workspace.

This layered publication resolves a false choice between radical transparency and traditional opacity. A paper asks the reader to trust that omitted work was performed appropriately. A raw workspace dump exposes too much and still leaves the reader with an unstructured burden. The living research book aims for selective epistemic disclosure: enough organised material to evaluate the claims, with explicit

statements about what is withheld, why, and under what conditions it can be inspected.

The model changes the meaning of completeness. A public book is complete not when it contains every event, but when it contains or points to everything necessary for its stated claims, subject to justified access restrictions. An internal workspace can be exhaustive and still scientifically unintelligible. A curated book can omit ninety percent of the history and be more accountable because the retained ten percent is structured around evidential relevance.

A practical schema for the future object has three principal layers. The private research workspace contains raw and evolving work. The controlled evidence vault contains artefacts required for audit, confidential replication, legal verification, or restricted review. The public living research book contains the coherent account, open evidence, disclosed limitations, and interfaces intended for the community. The boundaries are policy-governed and versioned. A claim in the public layer may cite an evidence object in the controlled layer and state the conditions under which qualified reviewers can inspect it.

This is not unprecedented in science. Clinical, security, defence, industrial, and privacy-sensitive research already uses controlled access. Patent systems exchange disclosure for time-limited exclusivity. Peer review already depends on confidentiality. The novelty lies in integrating these practices into the publication architecture rather than treating them as exceptions managed through email and institutional memory.

By 2040, mature systems should be able to compile these layers automatically enough that curation does not overwhelm researchers. Agents will classify information, propose redactions, detect personal data, identify possible inventions, verify licences, check whether claims have sufficient evidence, and generate explanations. Human governance will approve high-risk decisions. The quality of the system will depend less on fluent drafting than on whether these gates are reliable.

The synthesis of atomic and monographic futures yields the central wager of this book. Science will not choose between tiny publications and enormous ones. It will represent knowledge at both scales. Machines will operate over granular objects.

Humans will encounter integrated worlds. The publication institution that connects the two will be the living research book.

Table 2. Two scales of the same publication architecture.

Dimension	Implication for the living research book
Atomic substrate	Claims, evidence, experiments, data, software, literature relations, contributions, versions, and access policies receive independent identities.
Monographic interface	The programme accepts responsibility for a coherent explanation of why those objects matter and how they relate.
Article	A compact release note, priority record, or certification event attached to a larger research world.
Generated views	Briefings, papers, funder reports, tutorials, videos, simulations, and regulatory views compiled from a common source.
Assurance	Traceability connects each compression or expansion to stable source units and a declared edition.

3. LIVING BOOKS: FROM DIGITAL TEXTS TO AI-MAINTAINED KNOWLEDGE SYSTEMS

3.1 A tradition that already exists, but in fragments

The phrase living book predates the current wave of generative artificial intelligence and carries several intellectual traditions. In the humanities, it has named experiments that challenge the fixed, bounded, single-author monograph. Janneke Adema's Living Books argues for scholarly books as collaborative, processual, fluid, and open to remix rather than as finished commodities detached from the relations that produced them (Adema, 2021). Open Humanities Press and related experiments have treated publishing as an ongoing practice rather than a terminal event. These projects are not prototypes of an AI-generated technical manual, and they should not be appropriated as though they were. Their importance is deeper: they make the fixity of the scholarly book visible as a historical and institutional choice.

A second tradition appears in living evidence. Living systematic reviews are continually updated as new evidence becomes available (Elliott et al., 2017; Cochrane, 2026). Cochrane guidance treats this not as ordinary revision but as a distinct production and publication process, involving planned searches, decisions about when updates matter, workload, authorship, peer review, copy-editing, and eventual transition out of living mode. The lesson is crucial. A document does not become living because someone can edit a web page. It becomes living when maintenance, review, versioning, responsibility, and retirement are institutionalised.

A third tradition is the executable book. Jupyter Book and related systems combine narrative, code, data, equations, and interactive outputs (Executable Books Community, 2020). They demonstrate that a book can be read and run, that a figure can be connected to its generating computation, and that educational or

scientific narratives can include live artefacts. Executability is not universal reproducibility, but it narrows the distance between claim and procedure.

A fourth tradition is the research object. Rather than treating the article as the sole carrier of meaning, research-object approaches package data, workflows, software, people, provenance, and contextual metadata. RO-Crate is especially relevant because it provides a lightweight, extensible packaging mechanism intended for both human and machine use (Soiland-Reyes et al., 2022; Research Object Community, 2025). It does not prescribe the full epistemology of a research programme, but it supplies an important substrate for organising heterogeneous artefacts.

A fifth tradition is modular and granular publishing. Nanopublications, Octopus, ResearchEquals, registered reports, data papers, software papers, and protocol publications each detach a component of research from the final narrative article. They recognise that a method, dataset, hypothesis, review, or negative result can have independent value. They also expose a persistent problem: once the record is decomposed, somebody or something must integrate it.

A sixth tradition appears in collaborative knowledge systems such as wikis. Their strength is continuous improvement and distributed authorship. Their weakness, in the scientific context, is often the absence of stable versions, explicit evidential responsibility, domain-specific review, and clear relations between claims and primary artefacts. A wiki can explain a field, but it is not automatically the accountable publication of a particular research programme.

The living research book proposed here combines elements from all six traditions while changing their centre of gravity. It is collaborative and revisable like the living scholarly book, continuously updated like living evidence, executable like a computational notebook, packaged like a research object, granular like modular publishing, and navigable like a wiki. Yet it is governed as a programme-level scientific record. Its purpose is not only to remain open to change but to make change attributable, reviewable, and citable.

The distinction matters because the term living is easy to romanticise. Continuous change can undermine trust. If a reader cites a chapter and the chapter changes the next day, what was cited? If an agent rewrites an explanation after new evidence appears, who is responsible for the previous wording? If a result is

corrected, should the old version disappear? If a book has hundreds of contributors and automated processes, who can authorise a release? The printed book solved these questions through fixity. The living book must solve them through version governance.

Recent discussion of living publications in scientific journals has begun to frame them as interactive, AI-enhanced ecosystems that evolve with evidence (Oettl et al., 2026). This is close to the proposal developed here, but the programme-level scope adds several demands. The object must preserve not only an evolving review or article, but the concepts, methods, experiments, tools, data, and institutional decisions of a sustained line of inquiry. It must explain the relationship between multiple publications and show how the programme's own understanding changed.

The living research book is therefore not a digital book with frequent editions. Nor is it a chatbot trained on a PDF. It is a maintained epistemic institution. The prose, interfaces, agents, datasets, and simulations are components. The institution is the set of rules that determine what counts as an accepted claim, how versions are frozen, how corrections propagate, who receives credit, what remains confidential, and how readers can inspect the basis of an answer.

This interpretation preserves an important insight from experimental book scholarship: publishing is not the final packaging of knowledge after the real work has ended. It participates in the creation of the object. When a programme must explain its assumptions, connect claims to evidence, expose unresolved disagreements, and preserve the history of revisions, it may discover conceptual gaps that would remain invisible in a sequence of papers. The living book is not only communication infrastructure. It is an instrument of research.

3.2 The book as a versioned epistemic institution

A living research book must be alive in two distinct senses. It changes as knowledge changes, and it supports active interaction with that knowledge. These senses are related but should not be confused. A frequently updated PDF is dynamic but not interactive. A chatbot over a static corpus is interactive but not epistemically living. The proposed object combines continuous maintenance, stable editions, and queryable interfaces.

The canonical structure has a live branch and frozen releases. The live branch receives approved changes: new literature, experimental results, corrections, refined definitions, improved tutorials, altered confidence assessments, and updated interfaces. At defined intervals, commonly annual or biennial for large programmes, an editorial and scientific process freezes an edition. The edition receives a persistent identifier, an immutable content manifest, a model and software manifest, a contribution record, and a statement of unresolved issues. It becomes citable as a historical object even after the live branch moves on.

This duality resolves the apparent conflict between currency and stability. Living systematic reviews already face the problem that evidence must remain current while readers need to know when a version was assessed. Software releases face a parallel problem: development continues, but users depend on stable versions. Scientific books can adopt similar discipline without pretending that scientific truth behaves exactly like code.

Versioning must occur below the level of the whole book. A claim may change status without the surrounding chapter being rewritten. A dataset may receive a corrected release. A model may be retrained. A protocol may be clarified. A literature-positioning statement may be weakened because a previously overlooked paper is found. Stable identifiers for claims, figures, experiments, and artefacts allow citations to remain precise across editions.

The version history should not be a raw stream of edits. Scientific readers need semantic change descriptions. An edition should state which claims are new, strengthened, weakened, contradicted, retired, or merely re-explained. A correction that changes a conclusion is different from a stylistic improvement. A new tutorial should not trigger the same alarm as a revised data analysis. The publication system must classify changes and expose their significance.

Responsibility also becomes versioned. A contributor who approves a claim in one edition may not approve its later reformulation. An AI agent may draft a section using a particular model and retrieval corpus. The provenance record should identify the model family, configuration, source edition, and human approval associated with materially generated content. This does not mean publishing every prompt. It means preserving enough information to audit how authoritative text entered the book.

The governance body for a major living research book will resemble a combination of editorial board, release management team, and scientific steering group. Its role is not to centralise every edit. Routine corrections and generated explanations can follow delegated workflows. High-impact changes require stronger review: claims of novelty, alterations to principal conclusions, release of protected data, changes affecting patents, or revisions to ethical interpretation.

A living institution also needs an explicit end state. Not every book should be maintained forever. A programme may close, merge into another, or become historically stable. Living systematic review practice includes the possibility of leaving living mode when the question no longer changes rapidly or resources are no longer justified. The same principle applies here. A retired living book remains preserved, queryable, and citable, but it no longer implies current surveillance of the field.

The institutional character becomes especially visible in disputes. Suppose two teams within a programme disagree about the interpretation of a result. A conventional paper tends to force a single narrative or exclude the minority view. A wiki may host a discussion but leave the status ambiguous. A living research book can represent the disagreement explicitly. The canonical text can state what is agreed, what remains disputed, which evidence each interpretation relies on, and what observation would discriminate between them. Competing branches may be maintained until the issue is resolved.

This pluralism should not be confused with refusal to conclude. Scientific communication needs judgments. The institution must indicate the current accepted position and the confidence attached to it. But acceptance need not erase dissent. The history of rejected interpretations is often important for later reassessment and for understanding why a programme took a particular path.

The living book also makes correction less catastrophic and more normal. In paper culture, a correction can appear as a stain on a supposedly final object. In a maintained system, correction is an expected operation. The ethical requirement shifts from pretending that stable documents are infallible to ensuring that errors are detected, propagated, and preserved transparently. A claim that was once accepted and later rejected remains visible in historical editions, with clear warnings in current interfaces.

Citation practices must adapt. A citation to the whole book may be appropriate for general background. A precise scientific argument should cite the edition and the claim, section, dataset, or experiment. AI-generated answers should cite source units and disclose the edition used. When a claim changes status, downstream systems should receive machine-readable notices. The citation graph becomes a dependency graph that can support impact analysis: which conclusions rely on a result that has been corrected?

Libraries become central rather than peripheral in this model. They preserve frozen editions, manifests, and executable components. They may run trusted interfaces over institutional copies, reducing dependence on proprietary platforms. Their expertise in identity, metadata, licensing, preservation, and access control becomes part of scientific infrastructure. The living book does not abolish archiving; it makes archiving more demanding.

3.3 Beyond text and the end of compulsory linear reading

The phrase book may appear too narrow for an object that contains agents, code, and simulations. Its value lies in naming the canonical explanatory layer, not in limiting the media. The living research book is a book because it assumes responsibility for coherent exposition. It is a research environment because exposition is only one interface.

The conversational interface will be the most visible. A reader asks questions in ordinary language and receives answers grounded in the book's current or selected edition. The assistant can adapt to expertise, language, available time, and purpose. It can explain a result to a domain expert, a neighbouring discipline, a doctoral student, an investor, or a regulator without rewriting the canonical source for each audience.

A trustworthy research companion differs from a generic chatbot in several ways. It is bounded to an identified source and edition unless it explicitly searches outside. It cites the passages, claims, and artefacts used. It distinguishes the programme's position from external commentary. It exposes uncertainty and disagreement. It refuses to infer protected content from public answers. It records significant interactions when they become part of review or project governance, subject to privacy policy.

The companion should also know when a question cannot be answered from the public book. It can state that relevant evidence exists in a controlled vault, that the claim depends on licensed data, or that the information was withheld pending patent filing. This is more informative than either fabricating an answer or exposing protected material.

Structured conversations among AI systems provide a second interface. One agent can defend the programme's interpretation, another can represent a sceptical reviewer, a third can search for contradictory literature, a fourth can examine statistical assumptions, and a fifth can audit privacy or IP risk. Their exchange can be generated on demand or maintained as part of a review record. The point is not theatrical debate. It is to externalise lines of criticism that a single fluent answer tends to smooth away.

Such debates require strong controls. Agents can amplify shared model biases, produce the appearance of pluralism while drawing from the same training distribution, or generate endless low-value objections. Role labels are not evidence of independence. A credible system should vary retrieval sources, analytical methods, and model families where possible, and it should evaluate the debates by whether they identify testable issues rather than by rhetorical balance.

Executable interfaces connect claims to operations. A reader can rerun a figure, alter a parameter, inspect sensitivity, or execute a reduced experiment. In fields where full reproduction is too expensive or legally constrained, the book can provide validated surrogates, synthetic examples, partial workflows, or remote execution under controlled conditions. The result is not a binary label of reproducible or not reproducible. It is a map of what can be repeated, by whom, under which resources and permissions.

Simulations add another level. A scientific concept may be better understood through an interactive model than through prose. A reader can explore boundary conditions, observe failure cases, or compare competing theories. Generated simulations must be clearly separated from empirical evidence. A persuasive visualisation of a model is not validation of the model. The book should label whether a simulation reproduces measured data, illustrates a mechanism, or merely supports intuition.

Tutorials become dynamic derivatives. Instead of a single supplementary guide, the system can generate a learning sequence based on the reader's background. It can identify prerequisite gaps, create exercises, provide formative feedback, and connect every explanation to the canonical text. For a complex programme, this function may be as important as the publication itself. Research becomes reusable only when new people can understand and operate it.

Audiovisual material is generated from the same source. A five-minute overview, a detailed lecture, a laboratory demonstration, or an animated conceptual model can be produced in multiple languages and updated when the book changes. The system should preserve a link between each scene or statement and the source edition. Otherwise audiovisual outputs become untraceable publicity artefacts.

The book may also support spatial or immersive interfaces in domains where they add value. A molecular structure, archaeological site, physical apparatus, or network process can be explored interactively. The principle remains the same: every derivative experience must declare its relation to evidence and distinguish observation from reconstruction.

These interfaces transform peer review. A reviewer can ask the companion to produce a claim map, execute selected analyses, generate counterexamples, compare the programme's literature review with an independent retrieval, and stage a debate over the weakest inference. The reviewer's report can point directly to objects rather than page numbers alone. The system can then track how each issue was resolved across editions.

They transform collaboration as well. A team member can query why a design decision was made, see the evidence available at the time, and inspect later revisions. Meetings can update the research object through approved summaries rather than disappearing into unsearchable recordings. Coding agents can maintain executable components and tests. Research agents can monitor literature and propose changes. The book becomes the shared cognitive surface of the team.

The interfaces also create new accessibility opportunities. Explanations can be translated, simplified without losing traceability, converted to speech, or represented visually. These capabilities do not remove the need for accessible design and human evaluation. Generated accessibility can fail subtly. But a maintained multimodal source makes adaptation more feasible than a static PDF.

The danger is interface substitution. Users may trust the chatbot and never inspect the book. They may experience a smooth, personalised explanation that omits inconvenient details. The canonical text may become ceremonial while the real epistemic power resides in proprietary models. The design response is not to reject conversation but to make it accountable. Answers need source links. Important omissions need disclosure. Alternative views need availability. The book must remain independently inspectable and exportable.

The living research book is therefore not a collection of fashionable media. It is a hierarchy of representations tied to a common source. Text provides durable argument. Structured objects provide addressability. Code and data provide operational connection. Simulations provide exploration. Tutorials provide learning. Audiovisual forms provide communication. Agents provide navigation and criticism. Their unity comes from provenance and version control.

The extension from multiple media to a very large canonical source raises a final question: whether scale recreates the information overload that the system is meant to solve.

The image of a thousand-page research book is deliberately provocative. It seems to reproduce the very information overload that AI is expected to solve. The objection is valid if the book is understood as a conventional manuscript. It becomes weaker when the book is understood as a complete but selectively consumed scientific account.

A thousand pages may be appropriate for a decade-long research programme that spans theory, system design, experiments, regulation, deployment, and critique. Today, those pages already exist in dispersed form: papers, theses, deliverables, technical reports, repositories, meeting documents, websites, and private notes. The living book does not necessarily create more intellectual material. It integrates and maintains material that is already fragmented.

The decisive change is that linear reading becomes one option among many. A reader can traverse the canonical chapter sequence, but most interactions begin with a question. The system constructs a route. The route may include five pages from one chapter, a definition from another, an experiment, a simulation, and a recorded dispute. The reader sees why each component was selected and can expand outward.

This approach resembles a well-designed city more than a scroll. The city can be large because nobody is expected to walk every street to reach a destination. Size becomes problematic when there is no map, when roads are duplicated, or when important districts are inaccessible. The publication task is therefore architectural. The system must maintain conceptual maps, prerequisite relations, summaries at multiple resolutions, and reliable identifiers.

Progressive disclosure is the governing principle. The first view presents the contribution delta. The second explains the argument. The third exposes claims and evidence. The fourth opens methods, data, and executable components. The fifth may require controlled access. Each transition is available but not imposed. This preserves brevity without making omission irreversible.

A large book also supports different temporal views. A reader can ask what the programme believed in 2035, what changed in 2037, and why the 2040 edition rejected an earlier conclusion. Historical reconstruction becomes part of scientific understanding. This is especially valuable when models, datasets, and social conditions evolve rapidly.

The book should not aspire to self-sufficiency in the absurd sense of reproducing all prior science. It must still cite external work and rely on disciplinary knowledge. Its obligation is more specific: prerequisites and literature that materially affect the programme's claims should be explained enough that readers can understand the dependence without performing unnecessary archaeology. References remain paths to primary sources, not escape hatches from explanation.

The thousand-page form also changes writing quality. AI can generate connective prose, but integration requires editorial judgment. Repetition across chapters must be managed. Definitions must remain consistent. Contradictions must be surfaced rather than smoothed. Background should be allocated according to dependency, not according to what a model can easily produce. The living book needs automated coherence checks and human intellectual editors.

Coherence, however, is not always a virtue. A programme may contain genuine tensions. The system should not force all chapters into a single voice if that would conceal disciplinary differences or unresolved disagreements. The best living book is coherent about where it is incoherent. It explains the boundaries between perspectives and the evidence required to move them.

The large scale also invites strategic abuse. A programme may bury a weak novelty claim under exhaustive background. A reviewer may be intimidated by apparent completeness. Generated text may create a prestige effect: the object looks like an encyclopedia and is therefore treated as authoritative. The publication must counter this by making novelty, evidence quality, replication status, and unresolved risks visible independently of length.

A practical edition should begin with a contribution map and an epistemic status report. These are not marketing summaries. They identify the programme's major claims, confidence, supporting evidence, contradictory findings, dependence on protected material, and changes since the previous edition. A reviewer can assess the core before exploring the cathedral.

The thousand-page book is thus not a demand that science become verbose. It is permission for science to retain the context that compression once forced it to discard. AI changes the feasible relationship between completeness and attention. The challenge is to use that freedom for intelligibility rather than volume.

CHAPTER 4

4. THE ARCHITECTURE OF THE LIVING RESEARCH BOOK

4.1 Three publication zones: workspace, evidence vault, and public book

The most important architectural decision is also the one most likely to be misunderstood. The living research book is not a public mirror of the project workspace. A serious programme needs a rich internal environment in which people and agents can explore, fail, speculate, negotiate, and handle protected material. Publishing that environment wholesale would not produce transparency. It would produce an uncurated mass of information, violate legitimate confidentiality, expose personal and commercial interests, and burden reviewers with details that do not help them evaluate the claims.

The proposed system therefore separates three publication zones. The private research workspace is where the programme actually works. The controlled evidence vault preserves material that may be needed for audit, confidential review, reproducibility, legal verification, or future reinterpretation but cannot be released generally. The public living research book presents the coherent scientific account and the evidence that can be made openly available. These zones are linked by provenance and policy, not by the assumption that every object must move outward.

The private workspace is heterogeneous by design. It contains project plans, literature collections, raw data, notebooks, code branches, agent traces, meeting records, early hypotheses, failed experiments, legal advice, partner communications, security details, internal reviews, contribution records, and potentially sensitive reflections. It may include information that is wrong, offensive, provisional, or irrelevant. It is the space in which uncertainty can exist before it is made presentable.

A healthy research environment requires such a space. If every tentative idea is treated as an eventual public statement, people become less willing to explore. If every internal disagreement is automatically preserved for public scrutiny, teams may replace candid debate with defensive performance. If every raw model output is considered part of the scientific record, the record becomes unusable. Scientific accountability is not equivalent to continuous public surveillance.

The controlled evidence vault has a different purpose. It contains artefacts that matter to verification but require restricted access. Examples include pseudonymised or identifiable research data, licensed datasets, detailed security configurations, proprietary partner information, pre-filing invention material, high-cost models, complete audit logs, confidential ethics documentation, and computational environments that cannot be released openly. Access is granted by role, purpose, time, and legal basis. Reviewers may inspect selected objects inside a secure environment without downloading them. Auditors may receive broader access than public readers. Some evidence may be available only through approved queries or remote execution.

The public living research book is a curated publication, potentially extending to a thousand pages or more, but it is still a selection. It includes the prerequisite explanations, literature positioning, claims, methods, results, limitations, contribution records, open evidence, and provenance necessary for an informed public evaluation. It states when a claim relies on restricted material and describes the conditions under which that material was assessed. It may contain privacy-preserving aggregates, synthetic examples, redacted protocols, or attestations in place of raw data.

The relationship among the zones can be summarised as a publication contract. The workspace is optimised for discovery. The evidence vault is optimised for controlled verification. The public book is optimised for explanation and accountability. The same object may have representations in all three. Raw interview data remain private or controlled; a coded dataset may be available to qualified reviewers; an aggregate table and methodological explanation appear in the public edition. A source-code repository may remain private while a reproducible binary, test suite, interface specification, and third-party audit are published.

The boundary is not fixed once. A patent filing may allow previously protected technical detail to enter the next public edition. A data-use agreement may expire or be renegotiated. A participant may withdraw consent where the legal framework permits. A security vulnerability may be disclosed after remediation. Policy changes are therefore versioned events. The book should record not only what is public but why its visibility changed.

This architecture embodies the principle of selective epistemic disclosure. Information is disclosed according to its relevance to evaluating a claim and the legitimate constraints governing it. Selective disclosure is scientifically acceptable only when it is explicit. A project must not conceal weak evidence under the language of confidentiality. It should state what class of evidence is restricted, why access is limited, who has reviewed it, and what uncertainty the restriction creates.

The architecture also allows the public book to be more complete than a conventional article without demanding total exposure. A thousand pages can explain concepts, methods, assumptions, and limitations in depth while still excluding the complete chronological history. The public does not need every abandoned prompt, personal message, procurement discussion, or speculative hypothesis. It needs a defensible account of how the accepted claims relate to evidence.

The distinction is especially important for AI-generated work. Agent traces can be enormous. Preserving every intermediate token would be expensive and usually unhelpful. The publication system should retain decisions, source references, tool calls, model identities, relevant prompts or policies, and verification outcomes at a level appropriate to risk. High-stakes results may require more detailed trace retention than routine editorial transformations. The aim is auditability, not ritual logging.

The three-zone architecture should be implemented as a continuum of access policies rather than three physical folders. Objects can be encrypted, tagged, replicated, and governed across institutions. What matters is that the publication compiler can determine which representation is authorised for which audience and can prove the derivation of public claims without silently crossing boundaries.

Table 3. The three publication zones.

Dimension	Implication for the living research book
Private research workspace	Optimised for discovery. It contains raw work, failed paths, internal deliberation, protected data, code branches, partner material, and provisional agent outputs.
Controlled evidence vault	Optimised for restricted verification. It holds evidence required for audit, confidential review, remote execution, legal checks, and future reinterpretation.
Public living research book	Optimised for explanation and accountability. It contains the coherent argument, open evidence, disclosed limitations, provenance, and public interfaces.
Boundary rule	Objects move or are transformed across zones only under explicit privacy, security, licensing, contractual, and IP policies.

4.2 The claim-evidence-provenance graph

A long book alone does not solve scientific fragmentation. Its internal structure must make the relations among claims, evidence, methods, and sources explicit. The central technical object is therefore a claim-evidence-provenance graph, even if most users never see it directly.

A claim in this graph is more than a sentence. It has an identifier, formulation, scope, status, confidence, owners, date, version history, and relationships to other claims. It can support, contradict, refine, generalise, or depend on another claim. It can be designated as background knowledge, programme contribution, assumption, interpretation, forecast, or open question. These distinctions prevent a generated explanation from presenting every fluent sentence as an equivalent scientific assertion.

Evidence objects include observations, datasets, statistical analyses, formal proofs, simulations, benchmarks, qualitative materials, expert assessments, and replication reports. Each evidence object has provenance: who or what produced it, using which inputs, procedures, software, instruments, and environment. The graph records whether evidence is direct or indirect, public or restricted, independently reproduced or only internally validated.

Provenance is not merely a chain of file hashes. It connects technical lineage with scientific interpretation. A figure may be derived correctly from a dataset while the dataset is unsuitable for the claim. A model may reproduce an output while relying on an invalid assumption. The graph should therefore separate computational derivation from evidential support. The former can often be verified mechanically; the latter requires scientific judgment.

The graph also represents literature. External publications are not cited as undifferentiated documents. Relevant claims, methods, and evidence are linked to the programme's own objects. The system can show whether a source supplies background, supports a mechanism, reports a competing method, or weakens novelty. This makes literature positioning more inspectable and allows new publications to trigger targeted review.

Research objects and RO-Crate offer useful packaging principles for this substrate (Soiland-Reyes et al., 2022; Research Object Community, 2025). Persistent identifiers, contextual metadata, workflows, and provenance make artefacts machine-actionable. The living research book extends the approach by requiring an explicit argumentative layer. The crate tells the system what files and entities exist; the claim graph explains what scientific role they play.

The graph supports consistency checks. An accepted claim without evidence can be flagged. A figure generated from a superseded dataset can be identified. A conclusion that still cites a retracted source can trigger review. A new edition can be tested for regressions: did a change to a method invalidate a result reported elsewhere? Did a redaction remove the only public support for a claim? Did an updated definition alter the interpretation of older experiments?

It also supports uncertainty. Evidence does not simply switch a claim from false to true. The graph can record confidence ranges, domain restrictions, methodological weaknesses, contradictory observations, and dependence on contested assumptions. Different disciplines will need different evidence models. The architecture should not impose a universal numerical confidence score. It should make the programme's own standards explicit and machine-inspectable where possible.

The contribution graph intersects with the epistemic graph. People and agents are associated with conceptualisation, data curation, methodology, software,

investigation, validation, analysis, writing, review, and maintenance. CRediT provides an established high-level taxonomy, but the programme-level system can add granularity and temporal structure. A contribution is attached to an object or change, not merely to the book as a whole.

This enables a more honest account of AI participation. A model can be credited as a tool or agent for drafting, retrieval, code generation, or checking without being treated as a legal or moral author. Human responsibility remains attached to approval and scientific judgment. The graph can show where AI assistance was substantial and which controls were applied.

The graph must avoid becoming an ontology project that delays science indefinitely. Its purpose is operational. A minimal schema should cover the relations required for review, reproducibility, rights management, versioning, and credit. Domain-specific extensions can add richer semantics. The system should prefer usable metadata with declared gaps over a theoretically perfect representation that researchers cannot maintain.

AI reduces maintenance cost by proposing links and classifications. It can identify candidate claims in prose, map figures to generating code, detect references, suggest contribution roles, and flag missing provenance. But automatic extraction is fallible. The publication workflow should distinguish inferred metadata from confirmed metadata. High-impact relations require human or validated agent approval.

The graph is valuable even when the reader never encounters it. It powers the conversational assistant, review cockpit, generated tutorials, version comparisons, and targeted redactions. It allows the book to be large without becoming opaque. It is the structural reason that reversible compression is possible.

4.3 The epistemic publication compiler, editions, and citation

The compiler metaphor turns the architecture into a workflow. A software compiler transforms source code into an executable artefact under explicit rules. An epistemic publication compiler transforms a research workspace into public and controlled scholarly objects under scientific, legal, and editorial policies. The metaphor should not imply that judgment becomes fully deterministic. It highlights

that publication can be systematic, testable, and repeatable rather than assembled through ad hoc copying.

The compiler begins with a release intent. The programme decides that a new annual edition, a milestone report, a paper, a reviewer capsule, or a public tutorial should be produced. Each target has a profile defining required sections, evidence standards, access level, licences, contribution statements, and validation gates.

The compiler resolves dependencies. It identifies which claims and artefacts belong to the release, which versions they require, and whether any dependencies are missing or incompatible. It checks that major claims have evidence and that evidence has provenance. It verifies that references exist and that generated text is grounded in authorised sources. It detects unresolved contradictions and requires either resolution or explicit disclosure.

A policy engine then applies legal and institutional constraints. Personal data may be excluded, aggregated, anonymised, or placed in a controlled capsule. Licensed material may be described but not redistributed. Potential inventions trigger an IP gate before public disclosure. Security-sensitive detail may require redaction or delayed release. Partner agreements may constrain attribution or access. The system records these decisions and the authority under which they were made.

The editorial stage generates or updates prose. AI can draft explanations, integrate new evidence, harmonise terminology, identify prerequisites, and produce summaries at multiple resolutions. The draft is checked for citation fidelity, unsupported statements, contradictions, and changes in meaning. Human editors focus on intellectual structure, scientific emphasis, and the risks of false coherence.

The compiler produces multiple artefacts from the same release. The canonical public book is one. A concise paper or novelty note may be another. A funder report, public summary, data-management update, course module, video script, and reviewer environment can be generated simultaneously. Because they share source objects, inconsistencies can be detected automatically. A number reported in the public summary should not differ from the book unless the transformation is declared.

Validation is not a single pass. The system runs scientific and technical checks appropriate to the programme. Statistical analyses can be rerun. Code tests can

execute. Figures can be regenerated. Citations can be checked for support. Privacy leakage tests can be applied. Licence compatibility can be verified. Accessibility can be audited. A novelty agent can search for overlooked prior work, while a sceptical agent can identify assumptions that deserve human review.

No automated gate proves truth. The compiler produces an evidence package for release approval. Named humans or authorised committees accept responsibility. Their approval is itself a versioned object. This retains accountability while reducing routine labour.

The compiler also creates a release manifest. It identifies the edition, source objects, software and model versions, generated components, human approvals, licences, access restrictions, known limitations, and unresolved issues. The manifest allows libraries, reviewers, and downstream systems to inspect what the release contains without trusting a proprietary interface.

A public edition may include statements that certain evidence was reviewed under controlled conditions. The compiler can generate attestations without exposing the evidence. For example, an authorised data custodian may attest that an analysis was executed on a defined cohort and that reported aggregates match the output. A secure environment may provide cryptographic hashes or remote attestation. These mechanisms do not replace trust, but they make the trust relationship explicit.

The compiler supports corrections. When an error is discovered, the system identifies affected claims, sections, derivative outputs, and external citations where possible. It generates a correction candidate, re-runs validations, and publishes a new edition or patch. Historical versions remain preserved. A correction is thus propagated through the publication graph instead of attached as an isolated notice that readers may never see.

A mature compiler will be configurable across disciplines. A mathematical programme requires proof dependencies and formal verification options. A clinical programme requires consent, ethics, protocol registration, statistical analysis plans, and controlled data access. A qualitative programme requires contextual integrity, interpretive provenance, participant protection, and careful treatment of de-identification. The shared architecture supports profiles rather than enforcing one epistemology.

The compiler also creates a new object of research: the publication process itself. Its policies, validation accuracy, false positives, missed leaks, and editorial effects can be benchmarked. Teams can compare whether AI-assisted integration improves reproducibility or merely increases apparent completeness. The vision is scientific precisely because its infrastructure can be tested.

The compiler becomes a scientific institution only when its outputs remain stable enough to cite and flexible enough to change. This requires an explicit temporal architecture.

A living book without disciplined versioning becomes a moving target. A versioned book without readable change history becomes an archive that only machines can interpret. The publication architecture must therefore treat temporal structure as part of the scientific argument.

The live branch records the programme's current accepted state. It may change frequently, but not every change is immediately a public release. Routine updates can accumulate until validation thresholds are met. Urgent corrections can trigger a patch release. Major conceptual or evidential changes may justify a new edition.

Annual or biennial editions are plausible for large projects because they balance continuity with review capacity. The schedule should not be universal. A fast-moving public-health programme may need monthly updates. A mathematical programme may release only when proofs change. The essential feature is a declared cadence and criteria for exceptional release.

Each edition receives a persistent identifier and an immutable manifest. Chapters, claims, figures, datasets, and software versions can also receive stable identifiers. A citation should specify the level relevant to the argument. General orientation may cite the edition. A contested empirical claim should cite the claim and evidence version. A reproduced figure should cite the generating workflow.

Branching enables controlled pluralism. A programme can maintain an experimental interpretation without merging it into the canonical edition. Independent groups can fork a public book, propose alternative synthesis, or attach external replication results. The system records ancestry so that readers can see where interpretations diverged.

Forking raises governance questions. A scientific community should be able to criticise and reuse public material without allowing a misleading fork to impersonate the canonical programme. Persistent identity and provenance help. Licences determine what can be reused. Reputation and review determine what becomes credible. No technical mechanism can settle legitimacy alone.

Change logs must be intelligible. A machine-generated list of modified paragraphs is inadequate. The edition should include an epistemic delta: what new knowledge was added, what confidence changed, what errors were corrected, what methods were replaced, and what questions remain open. Readers should be able to request a domain-specific delta, such as changes relevant to privacy, clinical performance, or theoretical assumptions.

The public memory of error is especially important. Historical editions should not be silently altered. A current interface may place warnings on superseded claims and redirect readers to corrections, but the original record remains available. This preserves accountability and allows future historians or scientists to understand how the programme evolved.

Retractions become more granular. An entire book need not be withdrawn because one claim fails. The system can retire a claim, mark dependent results, and release a corrected edition. Conversely, if a foundational dataset is fraudulent, impact analysis may show that much of the book is compromised. Granularity enables proportionate response.

Citation metrics will also change. Counting citations to a thousand-page book tells little about which contributions were used. Claim- and artefact-level citations can improve attribution, but they can also create a new metric jungle. The system should preserve meaningful relations without assuming that every relation is a score. Responsible assessment principles remain necessary.

The temporal architecture supports project governance. Funders can compare the programme's promised objectives with the current edition and inspect changes over time. Reviewers can see whether earlier concerns were resolved. New team members can reconstruct the rationale behind decisions. Industrial partners can identify which public disclosures affect IP or contractual obligations.

The 2040 reading experience will therefore include a time dimension by default. A reader can ask not only "What does this programme claim?" but "When did it

begin to claim this, what evidence changed its confidence, who approved the change, and what downstream conclusions depend on it?" This is a richer form of scientific literacy than reading the latest PDF in isolation.

The architecture may appear elaborate compared with a paper. It is elaborate because the research already is. The existing system hides complexity by distributing it across informal practices, disconnected repositories, and human memory. The living research book makes the complexity explicit enough to govern.

5. RESEARCH AS CONTINUOUS KNOWLEDGE CONSTRUCTION

5.1 When research and writing cease to be separate phases

The conventional workflow distinguishes research from writing. The team investigates first and reports later. In practice the boundary is porous: researchers write notes, drafts, protocols, and code throughout the project. Yet the institutional workflow still treats the publication as a terminal transformation. A body of work is compressed into an article after the important decisions have already been made.

The living research book changes this relation. Writing becomes a continuous act of knowledge construction. This does not mean that scientists spend every day polishing chapters. It means that claims, definitions, evidence, decisions, and uncertainties are maintained in forms from which explanations can be generated. The publication is assembled throughout the research rather than reconstructed from memory at the end.

A project begins by representing its questions, assumptions, relevant literature, planned methods, ethical constraints, data sources, and expected evidence. These objects are provisional. As the work develops, experiments update claims, literature changes novelty assessments, and failures alter the path. The living book reflects this evolution. Early versions may contain explicit conjectures and unresolved alternatives. Later versions distinguish accepted results from abandoned directions.

This continuity improves memory. A late-stage author no longer needs to infer why a parameter was chosen two years earlier from an old notebook. The decision can be linked to the evidence and constraints available at the time. When the decision is revised, both states remain interpretable. New team members can understand the programme without relying entirely on oral transmission.

Continuous construction also enables earlier quality control. A claim without adequate evidence is visible before submission. A method that cannot be

reproduced from the recorded procedure triggers intervention while the researcher is still available. An undefined term that shifts meaning across work packages can be detected before it appears inconsistently in several papers. The book becomes a diagnostic surface for the programme.

AI agents are central because manual maintenance would otherwise become intolerable. They can extract candidate claims from notebooks, map outputs to experiments, monitor citations, suggest glossary updates, draft explanations, and identify gaps. They can ask researchers short, targeted questions when an important relation is unclear. The human burden should be concentrated on decisions, not transcription.

This requires careful interaction design. An agent that demands metadata after every action will be ignored. A useful system observes workflows, proposes structured updates, and allows lightweight approval. It distinguishes high-value events from noise. A failed exploratory run may require no durable record; a failure that changes the theoretical direction does. The system learns these thresholds through programme policy and human correction.

Continuous writing can also distort research. If the system constantly asks how an activity fits the book, researchers may optimise for narratability. Exploratory work that lacks immediate conceptual placement may be discouraged. The architecture must protect a pre-publication zone in which ideas can remain unstructured. Periodic curation is preferable to total real-time formalisation.

The distinction between raw chronology and scientific history is important. The living book should not simply replay events. It should construct an account of which events matter to the current understanding. This account can change. An experiment initially judged irrelevant may become important after a new theory emerges. Because the workspace preserves enough context, the programme can promote it into the durable record later.

The workflow therefore alternates between exploration and consolidation. During exploration, teams and agents generate possibilities. During consolidation, they identify claims, evidence, decisions, and implications. Each edition is a major consolidation point. This rhythm is compatible with creative research because it does not demand that every thought be immediately formalised.

The connection to registered reports is instructive. Registered reports separate the evaluation of questions and methods from the later results, reducing some incentives for outcome-driven storytelling. A living research book generalises the temporal transparency. It can preserve the planned analysis, record justified deviations, and connect each change to the evidence or constraints that motivated it. The aim is not to prohibit adaptation but to make adaptation legible.

In software-intensive research, continuous construction resembles continuous integration. Code changes trigger tests; data changes trigger validations; claim changes trigger consistency checks. A new result can be integrated only if required artefacts and explanations are present. The analogy has limits, but it suggests a mature engineering discipline for research records.

By 2040, a laboratory or research team may begin the day not with a blank document but with a programme environment that knows the current state of its theories, evidence, tasks, risks, and publication obligations. The environment does not decide what should be discovered. It makes the current knowledge state visible enough that humans can decide better.

5.2 AI agents as maintainers, critics, and institutional memory

The phrase AI research agent covers several distinct functions. Conflating them creates unrealistic expectations. A literature agent monitors external work and maps it to programme claims. A coding agent implements and tests computational components. A data agent validates lineage and quality. A methodological critic examines assumptions. A privacy agent checks data-use constraints. An IP agent identifies potential disclosures. An editorial agent maintains terminology and prose. A project-memory agent preserves decisions and resurrects relevant abandoned paths.

These agents need not be autonomous scientists. Their value can be high even when they operate as specialised assistants within governed workflows. The most credible near-term role is maintenance: tasks that are essential but difficult for humans to perform continuously at scale.

Literature monitoring illustrates the point. A living book contains an explicit map between its claims and external sources. An agent can search for new work,

classify relevance, identify potential contradictions, and propose updates. The programme's experts review high-impact findings. This is more useful than a generic weekly list because the search is connected to the programme's current epistemic structure.

The agent can also detect literature debt. A chapter may rely heavily on old sources in a rapidly changing field. A novelty claim may lack comparison with a newly popular method. A frequently cited assertion may trace back to a review rather than primary evidence. These signals do not determine correctness, but they focus human attention.

Coding agents maintain the executable side of the book. They can regenerate figures, run tests, update dependencies, and produce reproducibility packages. When an edition changes, they can verify that examples still work. Their outputs remain subject to review because generated code can contain subtle errors or security vulnerabilities.

Critic agents challenge coherence. One searches for internal contradictions. Another examines whether evidence supports the stated scope. A statistical critic checks power, assumptions, multiple comparisons, and uncertainty. A causality critic distinguishes predictive performance from causal explanation. A novelty critic looks for prior work. A red-team agent searches for adversarial failure cases.

The goal is not to produce a chorus of generic objections. Each critic should operate under an explicit method and report testable issues. Its performance can be evaluated against known errors and expert review. Agents that generate impressive language but low diagnostic value should be removed.

A particularly valuable role is institutional memory. Research programmes repeatedly lose knowledge when people leave, tools change, or attention shifts. The project-memory agent observes the evolving book and workspace. It can answer why a path was abandoned, which assumptions were never tested, and whether a new proposal resembles an earlier failed attempt. It can surface dormant ideas when new evidence makes them relevant.

This function must be governed carefully. Institutional memory can become institutional surveillance. Researchers should know what is retained, who can access it, and how long it persists. Personal performance discussions, sensitive interpersonal material, and speculative private notes should not automatically

become part of an agent's durable memory. The system needs boundaries between scientific rationale and personnel monitoring.

Agent collaboration can be represented as circuits or workflows rather than unstructured conversation. A literature update may pass through retrieval, relevance assessment, source verification, novelty impact analysis, editorial proposal, and human approval. A release may pass through evidence checks, privacy checks, IP checks, reproducibility tests, and editorial integration. Explicit orchestration makes responsibilities inspectable.

Models will change over the lifetime of a programme. The memory and publication architecture should not depend on one provider or model family. Agents operate over portable objects and record their outputs. When a model is replaced, the programme can compare results and re-run critical checks. This reduces the risk that a proprietary interface becomes the sole interpreter of the book.

Agent outputs also need epistemic status. A generated hypothesis is not a programme claim. A detected contradiction is not necessarily real. A proposed literature relation is not accepted until reviewed. The system should make these states visible. Otherwise machine suggestions can silently harden into canonical knowledge.

The strongest role for agents is therefore not authority but disciplined participation. They expand the programme's capacity to remember, compare, test, and explain. Humans retain responsibility for scientific judgment, ethical choices, and release decisions. The living research book provides the shared object around which this collaboration is organised.

5.3 Negative results, practical knowledge, and programme intelligence

Scientific publishing systematically loses negative and intermediate knowledge. Experiments that fail to support a hypothesis are less likely to become papers. Engineering approaches that fail under particular constraints may remain in issue trackers. Conceptual distinctions that were considered and rejected disappear from final narratives. Practical techniques remain tacit because they are too detailed for methods sections.

The loss has direct costs. Other teams repeat failed approaches. New members reopen closed debates. Meta-analysis overestimates effects when null findings remain unpublished. Historical understanding becomes distorted because successful paths appear inevitable. The living research book creates a place for these materials without requiring each to become a conventional publication.

A negative result can be represented at several levels. Some remain private because they are exploratory, low quality, or entangled with protected information. Some enter the controlled evidence vault because they affect the interpretation of public claims. Some become public because they close an important path or prevent duplication. The publication decision depends on epistemic value, quality, privacy, IP, and cost.

The book can integrate negative results narratively. A chapter can explain that three architectures were tested, two failed under defined conditions, and the third was selected. The underlying experiments can be inspected. This is more informative than a paper that presents the final architecture as if it had been obvious from the beginning.

Abandoned conceptual paths deserve similar treatment. A programme may reject a theory because it conflicts with evidence, because it is computationally intractable, because it depends on an unavailable dataset, or because priorities changed. These reasons have different scientific significance. Preserving them allows future researchers to distinguish a refuted idea from a postponed one.

AI-assisted memory can make abandoned paths productive. A new method may remove an old computational barrier. A newly available dataset may allow a previously impossible test. An agent can detect the relation and propose reopening the question. This capability turns historical documentation into a source of research opportunities.

Practical knowledge is harder. A written protocol can omit the judgment that an experienced researcher applies when conditions vary. Video, sensor traces, annotated examples, and interactive training can capture more of this knowledge. A living book can include demonstrations linked to specific steps and common failure modes. It can generate simulations or exercises that teach the decisions rather than merely the sequence.

The 2026 discussion of knowledge preservation in big science and AI is relevant because large collaborations face rapid turnover and complex infrastructures (Rainford et al., 2026). Documentation is often treated as overhead until an instrument, workflow, or person becomes unavailable. A living book makes preservation part of routine research, but only if maintenance is integrated into tools and recognised in careers.

Preservation has limits. Recording everything can create legal discovery risk, privacy exposure, and psychological inhibition. Some conversations are valuable precisely because they are temporary. The system should allow deliberate forgetting and retention schedules. Scientific provenance requires durability for accepted claims and evidence, not immortality for every message.

Deletion itself must be governed when it affects the record. If personal data are removed, the book may need to preserve an audit statement without retaining the data. If a security-sensitive trace is destroyed, the system records that the evidence can no longer be independently inspected. If a contributor requests removal of private material, the programme distinguishes personal content from scientific responsibility that must remain attributable.

The publication architecture therefore treats memory as selective and accountable. It preserves enough to support understanding, verification, and future work. It does not confuse completeness with indiscriminate accumulation.

The preservation of negative and practical knowledge changes the notebook itself. It becomes an input to a wider programme intelligence system rather than an isolated chronology.

The laboratory notebook has long served as a local record of observations, procedures, and ideas. Electronic notebooks improved search, sharing, and attachment of digital artefacts, but they generally remain organised around entries rather than the evolving structure of knowledge. The living research environment transforms the notebook into a programme intelligence system.

An entry still matters. A researcher records an observation, uploads data, or describes a decision. The system connects the entry to existing objects: an experiment, claim, risk, task, or literature source. It proposes new relations and asks for confirmation when significance is unclear. The notebook becomes one input channel into the programme graph.

The programme intelligence system maintains multiple horizons. It shows the immediate state of active work, the medium-term state of project objectives, and the long-term state of accepted knowledge. A result can move across these horizons as it matures. This prevents every preliminary observation from appearing as a public claim while preserving the path by which a claim emerged.

The system also integrates external obligations. A consent restriction follows the data. A software licence follows the dependency. A patent flag follows the invention. A deliverable requirement follows the project objective. These constraints are visible at the point of decision rather than discovered during final reporting.

Meetings become structured inputs. An agent may prepare a summary linked to decisions and claims, but participants approve what becomes durable. The full recording can be deleted according to policy while the accepted rationale remains. This reduces the accumulation of unsearchable meeting archives.

The system can generate a programme dashboard, but dashboards are not the main objective. A dashboard tends to simplify. Programme intelligence should allow movement from summary to evidence. It can show that an objective is "at risk," then explain which assumptions failed, which experiments are pending, and what decisions were taken.

The environment supports prospective review. Before a major experiment, agents can check whether the hypothesis is clear, whether the protocol matches the claim, whether the analysis plan is adequate, whether required approvals exist, and whether similar work has appeared. This does not eliminate error, but it moves scrutiny earlier.

It also supports reflection. At edition time, the programme can ask what it learned, which assumptions changed, where resources were wasted, and which capabilities were built. These questions are difficult to answer from a publication list. They are natural queries over a maintained research object.

The programme intelligence system should not become an automated manager that optimises researchers against narrow indicators. Its purpose is epistemic coordination. Operational metrics are useful, but they should remain subordinate to scientific goals. The same infrastructure that makes work visible can be used coercively. Governance must limit who can inspect individual-level traces and how they can be used in evaluation.

By 2040, the distinction between laboratory notebook, project management system, repository, publication platform, and research assistant may have weakened. The living research book is the public and reviewable surface of this integrated environment. The quality of the book depends on the quality of the work system beneath it, but the public does not receive that system in full. It receives a carefully compiled account.

6. TEAMS, AUTHORSHIP, AND CONTINUOUS PROJECT REPORTING

6.1 From author order to the long-lived research institution

Authorship performs several functions that have become increasingly difficult to reconcile. It attributes credit, assigns responsibility, signals participation, establishes reputation, and determines career value. The ordered list of names on a paper compresses these functions into a convention whose meaning varies by discipline. First authorship may indicate primary execution, senior authorship may appear last, alphabetical order may erase distinctions, and large consortia may contain hundreds or thousands of names.

CRedit improves this situation by identifying fourteen contributor roles, including conceptualisation, methodology, software, validation, data curation, investigation, supervision, and writing (Allen et al., 2014; CRedit, 2026). Its widespread adoption shows that the community recognises the inadequacy of author order. Yet a role statement attached to a finished article remains a retrospective summary. A living research programme can record contributions continuously and connect them to specific objects and changes.

A contribution ledger is not a cryptocurrency and need not be a public immutable log of every action. It is a versioned record of meaningful scholarly responsibility. A researcher can be associated with the design of an experiment, the curation of a dataset, the implementation of a method, the discovery of an error, the interpretation of a result, the maintenance of a benchmark, the writing of a chapter, or the approval of a release. The record can use CRedit-compatible categories while adding programme-specific detail.

This changes what can be recognised. A person who falsifies the team's preferred hypothesis may receive explicit credit for validation and conceptual correction. A software engineer who maintains the infrastructure for years is no

longer hidden in acknowledgements. A data steward's contribution remains visible even when multiple papers use the same dataset. A researcher who integrates several work packages into a coherent theory can be recognised for synthesis rather than only for prose.

The ledger also supports responsibility. If a claim is later corrected, the system can identify who produced, validated, and approved it. Responsibility is distributed rather than collapsed onto the corresponding author. This must be handled ethically. Attribution of responsibility should not become automatic blame. Scientific errors occur in complex systems, and correction is a positive contribution. The ledger should support accountability without creating a punitive culture that discourages risk.

Contribution records need consent and dispute procedures. People should be able to review how their work is represented. Teams need mechanisms for resolving disagreements about roles and significance. Automatic inference from activity logs can assist but should not decide. Number of commits, messages, or generated artefacts is not equivalent to intellectual contribution.

The system should also distinguish contribution from access. A person may have access to an object without contributing to it. A manager may approve a release without being entitled to authorship on every claim. An AI agent may generate substantial text or code without holding moral or legal authorship. The ledger records tool participation and human responsibility separately.

In the public book, contribution can be presented at several resolutions. The title page may name the programme and principal editors. A chapter lists its responsible contributors. A claim or artefact exposes a detailed record when needed. Career dossiers can extract verified contributions across programmes without forcing every contribution into a separate paper.

This model supports collective authorship. The programme can be the persistent authorial institution while individuals receive granular credit. This resembles how large technical systems, standards, or long-lived datasets are maintained, but the living book adds an explicit narrative and evidential account.

The model also challenges the heroic image of the solitary scientist. The point is not to deny individual creativity. It is to represent the infrastructure and collaboration that make many contemporary discoveries possible. A living book can show that a

conceptual insight depended on data engineering, experimental craft, criticism, and long-term maintenance. This richer account may influence how teams allocate resources and respect.

The danger is quantification. Once contributions are structured, institutions may assign points to each role. Researchers may optimise for visible ledger events and avoid invisible support. Managers may compare individuals using activity counts. The programme must therefore separate attribution from scoring. The ledger supplies evidence for qualitative evaluation; it should not become an automatic productivity ranking.

A second danger is strategic fragmentation. Researchers may claim separate micro-contributions to maximise credit. Governance should favour meaningful objects and collective outcomes. Contribution records are useful when they reveal relations, not when they create a market in intellectual tokens.

By 2040, authorship is likely to persist ceremonially and legally, but its evaluative monopoly can weaken. The living research book provides the infrastructure for a more accurate social contract: credit follows durable contributions, responsibility follows approvals and claims, and team science no longer has to pretend that a line of names describes the work.

The contribution ledger matters most when the team is understood not as the temporary author list of a paper but as an evolving scientific institution.

A paper has a temporary team. A programme has an evolving institution. People join, leave, change roles, form subgroups, and carry knowledge across projects. The living research book gives this institution a persistent intellectual surface.

The book becomes the common account against which members coordinate. It explains the programme's concepts, current claims, open questions, methods, and history. A new researcher can enter through a generated learning path. A collaborator can see where their expertise connects. A departing member can transfer responsibility for artefacts and unresolved issues.

This reduces dependence on a small number of memory holders. Many programmes have an informal centre: one person who understands the data pipeline, one who remembers why a method was rejected, one who knows the partner obligations. When such people leave, the programme loses more than

labour. The living environment aims to capture enough rationale and operational knowledge that continuity is possible.

The persistent book also allows teams to work at different temporal scales. A doctoral project may last three years, a grant four years, an infrastructure ten years, and a research theme several decades. Papers attach work to short episodes. A programme book can integrate contributions across funding cycles and preserve the lineage of concepts and tools.

This continuity changes collaboration between institutions. A consortium can maintain a shared canonical layer while partners retain private workspaces and controlled vaults. Provenance identifies which institution generated an artefact and which agreements govern it. The public book integrates results without requiring every partner to surrender all internal material.

Conflicts become visible at the right level. Partners may disagree about interpretation, ownership, release timing, or evidence quality. The system records decisions and unresolved positions. It can maintain a public consensus while preserving restricted documentation of the negotiation. This is preferable to either exposing confidential disputes or erasing their epistemic consequences.

The book can also support interdisciplinary work. Different fields use terms differently and accept different evidence. A maintained glossary and explicit concept relations reduce silent misunderstanding. Generated explanations can translate without pretending that the disciplines are equivalent. The programme can state that one claim is accepted under an engineering validation standard but remains preliminary under a clinical standard.

AI agents can serve as boundary objects. A domain agent explains concepts to another team, a compliance agent checks cross-border data use, and a methods agent maps evidence standards. Human experts remain necessary because translation can smooth important differences. The book provides the stable source against which translations are evaluated.

Leadership changes as well. A programme leader no longer controls the narrative simply by controlling the final paper. Contributions, disagreements, and evidence have durable representation. This can democratise teams, but it can also create new administrative power for those who control the publication infrastructure. Governance must distribute editorial authority and provide appeal mechanisms.

The long-lived institution needs succession. Editorial boards, data custodians, software maintainers, and claim owners cannot remain static. The book should record role transitions and ensure that critical objects have responsible stewards. An orphaned claim or dataset becomes a visible risk.

Funding boundaries create another challenge. A programme may continue after a grant ends, but maintenance requires resources. The institution must decide which components remain living, which are archived, and who pays. Libraries, infrastructures, funders, publishers, and research organisations may share this responsibility. Treating maintenance as a recognised contribution is necessary but not sufficient; it needs a budget.

The team-level effect is therefore both cultural and technical. Collaboration becomes less dependent on separate papers as territorial outputs. The programme's shared object makes integration visible and valuable. Individual careers can still extract verifiable contributions. The hoped-for result is not the disappearance of competition but a better alignment between what teams need and what researchers are rewarded for doing.

6.2 Reporting and annual editions as projections of maintained knowledge

Research projects currently report through a collection of documents and portals. A proposal states objectives, methods, milestones, risks, dissemination plans, and expected impacts. During execution, teams produce deliverables, periodic reports, milestone updates, publication lists, data-management plans, exploitation reports, ethics records, and public summaries. The same facts are rewritten in different formats, often by people far from the work.

Horizon Europe already uses continuous reporting for milestones, deliverables, publications, risks, and intellectual-property information, alongside periodic reporting (European Commission, 2026). CORDIS exposes project summaries, results, deliverables, and publications. These systems recognise that a project is more than a final report, but the information remains divided into administrative categories.

The living research book suggests a different architecture. Project reporting becomes a set of authorised projections from the programme's maintained object.

The objective map, evidence graph, contribution record, risk register, dissemination record, and exploitation status already exist in the environment. A report compiles the subset and explanation required by the funder.

This reduces duplication and improves consistency. A milestone marked complete must point to evidence. A deliverable claim must correspond to the public or controlled edition. A publication list is generated from release objects. A risk update is linked to the decisions and experiments that changed the assessment. The public summary is grounded in the same claims as the technical book.

The funder's interface can become queryable. An evaluator asks which objectives changed, why they changed, what evidence supports completion, which results depend on restricted data, and what unresolved issues threaten impact. The system generates a traceable answer. This is more informative than searching several PDFs and trusting that repeated text is consistent.

Reporting can also become less theatrical. Current project documents often encourage optimistic narratives because success must be presented periodically. A maintained evidence system can distinguish completed outputs from supported outcomes, and outputs from long-term impact. It can record that an objective was responsibly abandoned because its assumptions failed. A funder interested in learning should value such information, though political and administrative incentives may still punish it.

The proposal does not eliminate narrative reporting. Funders need interpretation, strategic context, and public accountability. It eliminates unnecessary re-authoring. The programme explains once at the canonical level and generates audience-specific reports with declared transformations.

Privacy and IP constraints are integrated. A public project page may describe a result without exposing proprietary details. A confidential reviewer receives a richer capsule. An exploitation officer sees invention and licensing status. The system prevents a public report from accidentally disclosing material flagged for patent review. It also prevents an internal confidentiality label from being used to hide all evidence of progress.

Continuous reporting becomes genuinely continuous because it is a view of ongoing work, not a portal updated before deadlines. Agents can monitor whether required information is missing and alert the team early. They can compare

commitments in the grant agreement with the current programme state. Human project managers focus on exceptions and negotiation.

This model affects proposal evaluation. A new project can inherit a research book from an earlier programme, showing not only publication counts but the actual state of knowledge, capabilities, negative results, and open questions. Evaluators can assess continuity and novelty more precisely. The proposal becomes a branch plan for extending an existing research world.

The risk is that funders demand access to the entire environment because it is technically available. The three-zone architecture must resist this. Contractual reporting should specify which objects and views are required. Continuous access does not justify unrestricted surveillance of researchers or partners. A programme can be accountable through compiled evidence without exposing every internal event.

Another risk is template capture. If the book is built primarily to satisfy funder categories, scientific structure will be distorted by administration. The canonical object must remain research-centred. Reporting profiles are derivative. Different funders should be able to receive their required views without forcing the underlying epistemology into one bureaucracy.

The most consequential projection is the annual or biennial edition, which serves simultaneously as a scientific release and an organisational milestone.

The annual or biennial edition is more than a publication schedule. It is a moment of collective reflection. The programme asks what it now knows, what changed, what failed, what deserves public release, what remains protected, and what should be pursued next.

The edition process begins with an epistemic inventory. Major claims are reviewed. Evidence status is updated. Literature positioning is rechecked. Reproducibility tests run. Negative results and abandoned paths are considered for inclusion. Contribution records are confirmed. Privacy, security, licensing, and IP gates are applied.

This process can replace some of the fragmentation created by multiple paper deadlines. Instead of each subgroup optimising its own narrative, the programme must integrate. Papers can still be produced, but their claims are reconciled with the

shared account. Contradictions become explicit before they reach the literature as unrelated publications.

The edition also supports external review. A programme may invite a panel to evaluate the annual change set rather than re-review the entire book. Reviewers inspect new and altered claims, unresolved issues, and selected foundational components. The resulting review becomes part of the next edition.

For a consortium, the edition is a governance event. Partners approve what is attributed, released, and protected. Disputes about ownership or interpretation are resolved or documented. Responsibilities for the next period are assigned. The book becomes a practical mechanism for alignment.

Annual editions can also communicate failure honestly. A programme can state that a central hypothesis was rejected, that a planned technology proved infeasible, or that a dataset was unusable. The edition explains what was learned and how the direction changed. This form of reporting is scientifically valuable but depends on funders and institutions not treating every deviation as incompetence.

The cadence creates a stable unit for careers. Researchers can cite their contributions to a specific edition. Promotion committees can review a coherent body of work and verified role records. The programme can demonstrate sustained maintenance and integration that paper counts ignore.

The risk is that the edition becomes another ceremonial deliverable. Agents generate a polished book, leaders approve it, and nobody uses it. Avoiding this requires that the book remain operationally connected to research, review, onboarding, reporting, and tool execution. A living book that does not affect decisions is merely an expensive publication.

The annual edition is therefore both release and test. If the programme cannot explain its current knowledge coherently, trace its important claims, and identify its unresolved problems, the difficulty reveals a research-management problem, not only an editorial one. The book makes the programme accountable to its own understanding.

7. PEER REVIEW OF A RESEARCH WORLD

7.1 The reviewer should judge science, not perform literature archaeology

Peer review has accumulated tasks that no small group of unpaid experts can perform reliably under current conditions. A reviewer is expected to assess novelty, methodological validity, statistical analysis, clarity, ethics, relevance, and relation to an expanding literature. The manuscript may depend on code, data, supplementary files, and specialised techniques outside any one person's expertise. Review invitations arrive faster than available attention.

The article makes the problem harder because it is a compact narrative surrounded by external dependencies. To judge a novelty claim, the reviewer searches literature. To inspect a method, the reviewer follows references. To understand a dataset, the reviewer opens another repository. To reproduce a figure, the reviewer reconstructs a computational environment. Much review time is spent recovering context that the publication system has distributed.

A living research book changes the reviewer's task. The reviewer is not asked to read a thousand pages or inspect the whole project workspace. The system creates a review route centred on the contribution under evaluation. It identifies the claims that are new or changed, the evidence they rely on, the literature positioning, the methods, the limitations, and the protected components available under controlled access.

The reviewer's most valuable work becomes judgment. Does the experiment actually test the claim? Is the interpretation proportionate to the evidence? Is the novelty intellectually meaningful or merely technical? Are alternative explanations taken seriously? Does the programme understand the limits of its method? These questions cannot be reduced to retrieval and formatting checks.

AI support should therefore remove clerical and archaeological burden rather than simulate final judgment. A reviewer assistant can retrieve relevant sections, compare terminology, check citations, regenerate figures, inspect statistical assumptions, and surface contradictory literature. It can produce a map of unresolved issues and show how similar concerns were handled in earlier editions. The reviewer can verify the assistant's path and challenge its source selection.

This division of labour responds to current publisher policies. Major publishers warn reviewers not to upload confidential manuscripts into public generative AI systems because of confidentiality, data protection, and intellectual-property risks (Nature Portfolio, 2026). They also insist that scholarly judgment and responsibility remain human. These restrictions are justified, but they leave reviewers with a practical demand for support. A programme-controlled or publisher-controlled review environment can supply AI assistance within the confidentiality boundary.

The system should not infer that a long book deserves a long review. Review scope is explicit. A new edition may contain hundreds of editorial improvements but only five materially changed scientific claims. The review focuses on those changes and on any dependencies they affect. Foundational components can be sampled or periodically re-audited.

The scope can also be distributed among reviewers. A methodological reviewer evaluates design and analysis. A domain reviewer evaluates significance and positioning. A privacy reviewer examines data governance. A software reviewer inspects reproducibility. Their reports target the same claim graph and can see where assessments overlap or conflict. The editor integrates judgments rather than receiving independent essays that refer vaguely to manuscript pages.

This architecture enables more precise review outcomes. A reviewer may accept a method, reject one interpretation, request stronger evidence for another claim, and approve publication of the rest. The book can release with explicit claim statuses rather than waiting for complete consensus. This is closer to how scientific knowledge actually matures.

Review also becomes educational. A programme can respond to a critique by updating the book, adding an experiment, clarifying a prerequisite, or preserving a disagreement. Future readers see the issue and its resolution. The review is not

only a hidden gate but part of the evolving scientific account, subject to confidentiality choices and reviewer consent.

The main risk is automation bias. A reviewer may trust the assistant's literature map, statistical flags, or summary and fail to inspect what was omitted. The system must make selection visible. It should allow independent search, competing agents, and manual navigation. A reviewer's use of AI should be recorded at an appropriate level so that the editor understands how the judgment was supported.

Another risk is the opposite: the assistant generates so many warnings that review becomes impossible. A good review environment prioritises issues by potential impact and provides evidence for each flag. Its performance is evaluated against expert findings. Generic criticism is not quality assurance.

7.2 The reviewer cockpit and the controlled review capsule

The reviewer cockpit is the practical interface to the living research book. It is not a dashboard that reduces science to traffic lights. It is a navigable environment that begins with a contribution map and allows the reviewer to descend into evidence.

The first view states the release scope. It identifies new claims, changed claims, corrected claims, new methods, affected dependencies, and unresolved disputes. It shows which components were generated or substantially transformed by AI and which validations were performed. It also states which evidence is public, controlled, unavailable, or independently attested.

The reviewer can select a claim and see its formulation, scope, confidence, supporting and opposing evidence, relevant literature, responsible contributors, and change history. A claim may depend on a restricted dataset. The interface explains the restriction and offers the authorised review mode, such as remote analysis, aggregate inspection, or third-party attestation.

The controlled review capsule contains only what is necessary for the assigned review. It may include protected data views, code, detailed protocols, confidential partner information, or pre-publication invention material. It is isolated from public AI services. Access is time-limited, logged, and governed by confidentiality obligations. Export may be restricted. The reviewer assistant operates within the capsule and does not reuse content for model training.

The capsule is preferable to full workspace access for both scientific and ethical reasons. The full workspace contains irrelevant noise and sensitive material. Reviewers should not inspect personnel discussions, unrelated commercial plans, raw identities, or speculative notes unless these are directly relevant and legally accessible. More access does not automatically produce better review.

The public book and review capsule should be consistent. The reviewer can verify that public claims accurately represent controlled evidence. The system highlights transformations, such as aggregation, redaction, or synthetic replacement. If the public explanation omits a limitation visible in the controlled evidence, the reviewer can require correction.

Remote execution is a key capability. A reviewer may rerun an analysis inside the controlled environment without receiving the underlying data. The system records the code version, environment, and output. The reviewer can vary parameters within authorised bounds. For computationally expensive or sensitive work, the programme may provide pre-defined tests or allow queries through a secure interface.

Cryptographic and confidential-computing techniques can strengthen the capsule. Hashes establish that a reviewed artefact corresponds to a preserved object. Remote attestation can provide evidence about the execution environment. Trusted execution environments can protect data in use from parts of the surrounding infrastructure. These mechanisms reduce some risks but do not establish scientific validity. The reviewer still needs to assess whether the data and analysis answer the question.

The cockpit should expose model mediation. If an AI summarises a protected dataset, the reviewer can inspect the query and source objects. If a model proposes a criticism, the interface identifies the method and evidence. The reviewer can disable generated assistance and navigate directly. This is important because protected environments can create false confidence: security of access does not imply quality of inference.

The capsule also supports blind and open review variants. Identities can be hidden at the interface level where feasible. Open-review comments can be published with the edition. Confidential comments can remain in the vault while their

resolutions are summarised publicly. The architecture does not prescribe one review norm; it makes the selected norm operational.

Review assignment can improve because the book's structure exposes required expertise. Editors can identify that a release needs expertise in causal inference, differential privacy, and a particular application domain. Conflicts of interest can be checked against contribution and collaboration graphs. The system should avoid fully automated assignment because social and intellectual conflicts are not always represented in metadata.

The reviewer cockpit can also show disagreement among automated checks. A novelty agent may identify substantial prior work while the programme's agent considers the contribution distinct. A statistical agent may flag an assumption that a domain method treats as standard. Presenting disagreement prevents the system from manufacturing a single artificial verdict.

Accessibility matters. A cockpit that requires reviewers to learn a complex ontology will fail. The default path must be intelligible, with structure available when needed. Reviewers should be able to write ordinary prose, attach comments to objects, and request explanations. The system translates these actions into the underlying graph.

The review capsule introduces costs. Secure infrastructure, access management, data custodians, and technical support are not free. The model is most justified for significant programmes and high-stakes claims. Small contributions may use lighter profiles. The architecture should scale in assurance rather than impose maximum complexity universally.

Table 4. Review access is broader than public access but narrower than workspace access.

Dimension	Implication for the living research book
Public reader	Receives the canonical book, open artefacts, assurance labels, and an explanation of restricted dependencies.
Assigned reviewer	Receives a scoped capsule containing the material needed for a defined methodological, domain, privacy, software, or IP review.
Independent auditor	May inspect broader protected evidence, release procedures, access logs, or attestations under a separate mandate.

Dimension	Implication for the living research book
Project team	Retains the full exploratory workspace, subject to internal governance and retention limits.
Prohibited inference	No role receives unrestricted access merely because more information exists; access follows purpose and necessity.

7.3 Continuous, claim-specific, adversarial, and visible review

A static article receives review at a single point, usually before publication. A living book changes continuously, so review must have a temporal model. Continuous review does not mean that reviewers monitor every edit. It means that material changes trigger appropriate scrutiny and that later evidence can update earlier judgments.

Claim-specific review is the first mechanism. A new or revised claim receives a review status independent of the whole book. The status can indicate internal validation, external review, replication, contestation, or retirement. Readers can filter the book according to the assurance level they require.

Periodic edition review is the second mechanism. An external panel evaluates the epistemic delta and samples stable foundations. This avoids re-reviewing everything while providing programme-level oversight. The panel can identify systemic problems that claim-level checks miss, such as a recurring bias in data collection or a narrowing theoretical frame.

Post-publication review is the third mechanism. External researchers can attach critiques, replications, or alternative analyses to specific objects. The programme responds in the live branch and includes material resolutions in the next edition. The system preserves disagreement rather than allowing comments to disappear in separate platforms.

Adversarial review adds structured attempts to break claims. Agents generate counterexamples, search for contradictory evidence, test sensitivity, inspect data leakage, and identify security or privacy failures. Human red teams can be commissioned for high-stakes results. The goal is not antagonism for its own sake but deliberate exposure to failure modes.

Adversarial methods need boundaries. A medical programme cannot expose participant data to arbitrary attacks. A cybersecurity result may require coordinated vulnerability disclosure. An industrial programme may need patent filing before public challenge. The review architecture schedules and contains adversarial access according to risk.

The quality of automated review must be measured. Research on AI-assisted peer review shows both promise and concern. Models can summarise and identify surface issues, but they may struggle with novelty, deep methodological flaws, and contextual judgment. They can also be manipulated by prompt injections hidden in manuscripts. A review environment must treat submitted content as untrusted input, separate instructions from data, and test agents against adversarial documents.

The system should maintain benchmarks of known review failures. Can the agent detect unsupported citations, leakage between train and test data, p-hacking indicators, inconsistent sample counts, or claims that exceed the evidence? Can it avoid false accusations? Performance should be reported by task and domain, not as a generic "AI reviewer score."

Human review quality also varies. The structured environment can support calibration. Editors can compare reviewer judgments, identify systematic disagreement, and provide feedback. Reviewers can see which concerns proved consequential in later editions. This creates a learning system rather than a sequence of isolated reports.

Continuous review changes the meaning of acceptance. Publication is not a permanent certificate of correctness. It is a statement that an object met defined standards at a particular edition. Later evidence can change status. This is already true in science; the living book makes it explicit.

The model can reduce review crisis only if it uses human attention selectively. Automated checks handle routine consistency and navigation. Reviewers focus on high-impact judgments. The system should compensate substantial review work and record it as a scholarly contribution. Otherwise better infrastructure will merely increase expectations placed on the same unpaid labour.

For continuous review to improve science rather than merely accelerate gatekeeping, its judgments and resolutions need a durable history.

Traditional peer review often disappears behind the published paper. Even when reports are open, they remain attached to one manuscript version. The living research book can integrate review into the public history of the programme without requiring every comment to be public.

Each significant issue has an identifier, target, author or agent, date, visibility level, and resolution. The resolution may include a textual change, new experiment, limitation, rejected request, or continuing disagreement. The system records the rationale.

This creates accountability for authors and reviewers. Authors cannot claim that a concern was addressed when the relevant claim remains unchanged. Reviewers can see the consequences of their comments. Editors can identify unresolved high-risk issues. Readers can inspect how the programme responded to criticism.

The history also protects legitimate disagreement. A programme may reject a reviewer's request because it rests on a different epistemic standard or would require disproportionate resources. The public edition can summarise the disagreement. Acceptance does not require pretending that all reviewers agreed.

Review histories can become valuable teaching material. Students can see how claims were challenged and refined. Methods courses can examine real examples of scope correction, failed replication, or interpretive dispute. This is more instructive than presenting finished papers as flawless products.

There are privacy and reputational risks. Reviewers may be less candid if every comment is public. Authors may suffer from the permanent visibility of preliminary errors. The architecture should support confidential, pseudonymous, delayed, and open review modes. Public summaries can preserve epistemic value without exposing unnecessary personal detail.

The system should also protect against review capture. Powerful programmes could curate the visible history to display only favourable criticism. Independent identifiers, editorial oversight, and preserved controlled records are necessary. External review platforms and libraries may maintain attestations that a published summary corresponds to the actual process.

By 2040, a reader encountering a major claim should be able to see not only the evidence but the history of serious objections. Scientific trust will come less from

the absence of visible conflict and more from evidence that conflict was handled competently.

The peer review of a research world is therefore not a larger version of manuscript review. It is an ongoing institution that allocates attention, controls access, records judgment, and updates assurance as the object changes. The public book provides coherence. The review system ensures that coherence does not become insulation from criticism.

8. SELECTIVE TRANSPARENCY: PRIVACY, CONFIDENTIALITY, AND INTELLECTUAL PROPERTY

8.1 Openness, privacy-preserving publication, and secure review

Open science is sometimes described through a moral contrast between openness and secrecy. The contrast is too simple for modern research. A programme can be publicly accountable while protecting personal data, participant confidentiality, trade secrets, security-sensitive information, and inventions before filing. It can also claim openness while hiding essential weaknesses behind inaccessible data or proprietary systems. The relevant question is not whether everything is open. It is whether access restrictions are justified, proportionate, explicit, and compatible with evaluation of the claims.

The living research book makes this question unavoidable because it can potentially integrate far more of the project than a paper. A private workspace may contain personal identifiers, behavioural traces, health records, confidential interviews, partner data, source code, model weights, threat analyses, contractual terms, and internal deliberations. Publishing a coherent thousand-page book does not imply publishing these materials. The book is an output of governed selection.

European data-protection law supplies a useful baseline. The General Data Protection Regulation requires purpose limitation, data minimisation, accuracy, storage limitation, integrity, confidentiality, and accountability (European Parliament and Council, 2016). Scientific research can benefit from specific legal provisions and derogations, but Article 89 requires appropriate safeguards, with technical and organisational measures that respect data-minimisation principles. The practical implication is clear: research status does not convert unrestricted data retention or publication into good practice.

The public book should therefore disclose the data logic rather than necessarily the data. It can explain what categories were processed, for which purposes, under which governance, with what sampling and exclusion decisions, and with what limitations. It can publish aggregates, anonymised derivatives, synthetic examples, schemas, code, and data dictionaries where appropriate. It can state what qualified reviewers were allowed to inspect and what remains unavailable.

Anonymisation is not a magic operation. Rich datasets can permit re-identification through linkage. Text, images, genomic data, mobility traces, and small subgroups can remain highly revealing. The publication compiler should treat privacy risk as contextual. Removing names is insufficient. A privacy review evaluates whether the released combination of variables, examples, model outputs, and auxiliary information creates an unacceptable risk.

Data minimisation applies to the book itself. An explanation should not include personal detail merely because the source workspace contains it. A generated example should not accidentally reproduce a participant's rare narrative. A video tutorial should not reveal information hidden in the text. A conversational agent must resist attempts to infer protected facts through repeated queries.

This last risk becomes significant in interactive publications. A static public table exposes a bounded output. A chatbot can answer many adaptive questions, allowing users to combine information and probe boundaries. The public companion needs a policy layer that limits access, detects extraction patterns, and ensures that generated answers remain within the approved disclosure surface. Its safety cannot rely solely on a system prompt.

Privacy also concerns researchers and reviewers. Internal conversations may contain personal opinions, performance discussions, or protected disclosures. Contribution records can become behavioural surveillance. Review logs can reveal identities or confidential judgments. The architecture should minimise person-level data, define retention, provide role-based access, and prevent scientific provenance from becoming a general employee-monitoring system.

Selective transparency must be symmetrical. The programme discloses restrictions and their epistemic consequences. A reader should know when evidence could not be independently accessed, when an analysis was performed only by the originating team, or when a third-party custodian supplied an attestation.

These facts affect confidence. Privacy protection is not a rhetorical shield against critique.

The ideal is verifiability without universal visibility. This can be achieved through layered access, independent custodians, secure computation, reproducible code, synthetic or aggregate data, and transparent statements of residual uncertainty. No mechanism eliminates the trust relationship, but the relationship can be specified and audited.

These principles become operational through privacy-aware classification, compilation, and controlled review.

The technical design begins with classification. Every object entering the research environment receives or inherits labels describing sensitivity, legal basis, consent, contractual restrictions, security risk, retention, licence, and intended audience. AI can propose classifications, but high-risk labels require validation. The labels travel with derived objects through provenance relations.

Classification enables policy-aware compilation. A public edition cannot include a restricted object directly. It may include an approved abstraction. A reviewer capsule can include a controlled view. An internal edition can include the full object. The compiler records the transformation and checks whether the derivative leaks information beyond the authorised purpose.

Access control should follow least privilege. A methodological reviewer receives the evidence needed for methodological review, not the consortium's entire protected repository. A data-protection auditor receives processing records and privacy analyses, not commercial negotiations. An IP reviewer receives invention-relevant material, not identifiable participant data. Roles can overlap, but access is granted intentionally.

Secure data environments allow reviewers to inspect or analyse protected data without receiving copies. Queries and code run inside the environment. Outputs are checked before export. The environment records the operations necessary for audit. For very sensitive work, independent data custodians can operate the system so that neither the public nor the publication platform controls the raw data.

Confidential computing can strengthen protection of data in use through hardware-based trusted execution environments and attestation (Confidential

Computing Consortium, 2026). It can reduce exposure to operating systems, hypervisors, or cloud administrators. This is relevant when AI agents or review tools process confidential material. It is not a complete solution. Trusted environments have implementation risks, side channels, supply-chain dependencies, and limited protection against authorised but malicious queries. They belong in a defence-in-depth strategy.

Differential privacy can be useful when publishing statistics or enabling repeated queries over sensitive data (Dwork and Roth, 2014). By bounding how much an individual's participation can affect an output, it provides a formal privacy guarantee under defined assumptions and parameters. Its suitability depends on the scientific question. Noise can make small effects or subgroups difficult to study, and privacy budgets can be misconfigured. The book should report the mechanism and its implications rather than presenting "differentially private" as a generic quality label.

Synthetic data can support tutorials, software testing, and some forms of reproducibility. It can reduce direct exposure, but it can also preserve sensitive patterns, omit important relationships, or create false confidence that results transfer to real data. The public book should distinguish findings validated on original data from demonstrations performed on synthetic substitutes.

Federated analysis allows computation across institutions without centralising raw data. Secure multiparty computation and homomorphic encryption can provide stronger cryptographic guarantees for selected operations. These techniques are valuable when the research design justifies their cost and limitations. The living book need not expose implementation detail to every reader, but its privacy section should explain what was protected, against which threat model, and with what residual risk.

The reviewer companion requires special controls. Current publisher policies prohibit reviewers from uploading unpublished manuscripts to public AI tools because manuscripts may contain confidential or proprietary information. A future system should provide a bounded model whose inputs and outputs remain within the review environment, whose retention and training policies are explicit, and whose tool access is restricted. Model providers should not gain secondary rights over the material.

Prompt injection must be treated as a security issue. Manuscripts, data, or retrieved web pages can contain instructions designed to manipulate agents. The review system should separate untrusted content from control instructions, restrict tool permissions, sanitise outputs, and maintain independent validation. A secure hosting environment is insufficient if the agent follows malicious content.

Privacy-preserving publication also needs query governance after release. A public assistant can inadvertently combine open facts into a sensitive inference. Rate limits, policy checks, output monitoring, and privacy testing are necessary. High-risk domains may provide a limited question-answering interface rather than unrestricted agent access.

The system should support rights of data subjects and participants where applicable. If data are corrected or withdrawn, provenance identifies affected analyses and editions. The programme can assess whether public aggregates need revision. Historical scientific records may need to preserve that a change occurred without retaining the removed personal data. These tensions require domain and jurisdiction-specific governance.

Privacy is thus not a chapter added to publication after the science is complete. It shapes what evidence can be collected, how it can be analysed, which agents can access it, what reviewers can inspect, and what the public book can claim. The living research environment makes these dependencies visible.

8.2 Patents, trade secrets, copyright, database rights, and licences

Intellectual property creates a different but related boundary. A research programme may have obligations to publish and incentives to protect commercially valuable results. Public disclosure can destroy patent novelty in jurisdictions operating under strict novelty rules. Trade-secret protection depends on information remaining secret and on reasonable steps to preserve confidentiality. Copyright protects expression rather than ideas, methods, or facts. Database rights and contractual licences can constrain reuse even when individual facts are not protected. The publication compiler must treat these regimes as distinct.

Patent strategy is especially sensitive to timing. The European Patent Office warns that public disclosure before filing can prevent a worthwhile patent, and the

state of the art broadly includes what has been made available to the public before the filing date (European Patent Office, 2026). Limited exceptions exist, but relying on them is risky. A living book that updates continuously could accidentally disclose an invention in a chapter, figure, generated answer, video, or public change log before the IP team has acted.

The architecture therefore needs an invention gate. Agents can flag potentially patentable technical contributions, but human IP professionals and inventors decide. Material under review remains in the private workspace or controlled vault. Public generation is blocked until a release decision is recorded. After filing, the programme can disclose according to strategy and contractual obligations.

The gate should not become an excuse for indefinite secrecy. A project funded under open-science obligations may need to disseminate results, and patents themselves rely on disclosure. The programme records the reason and expected duration of protection. Funders and partners can see that a result exists without necessarily seeing the enabling detail before filing.

Trade secrets require a different approach. Patents exchange disclosure for time-limited exclusive rights. Trade secrets can last while information remains confidential, but they do not prevent independent discovery. WIPO notes that technical information such as experimental data, algorithms, source code, processes, and combinations of public elements may qualify when they have commercial value, are known to a limited group, and are protected by reasonable steps (World Intellectual Property Organization, 2024, 2026).

A programme that relies on trade secrets may publish scientific findings while withholding implementation details. This creates an epistemic cost. If the secret details are necessary to reproduce the claim, public verification is limited. The book should state the limitation and provide alternative assurance where possible: independent audit, benchmark access through an interface, controlled review, formal specification, or test vectors.

Selective disclosure must not misrepresent a result as fully reproducible. A reader should know whether the public method is complete, whether a proprietary component is essential, and whether independent parties have validated it. The programme may decide that commercial protection justifies reduced openness, but the scientific record should not hide the trade-off.

Copyright in the living book is complex because the object is collaborative, continuously revised, and partly AI-generated. WIPO's basic distinction remains useful: copyright generally protects original expression, not the underlying ideas, procedures, or facts. Human authorship requirements for AI-generated material vary and remain contested. A programme should not assume that every generated passage enjoys the same protection as human-authored text.

The practical solution is rights metadata at the object level. Text, images, code, datasets, model outputs, and third-party quotations can carry authorship, licence, and source information. The compiler checks compatibility before release. A chapter may be open under a Creative Commons licence while code uses an open-source licence and a dataset remains under controlled terms.

Database rights matter particularly in Europe, where a database can receive copyright protection for original selection or arrangement and a *sui generis* right when substantial investment has been made in obtaining, verifying, or presenting contents. Contractual terms can further restrict access and reuse. A living book that integrates literature, datasets, and knowledge graphs must respect these layers rather than assuming that publicly accessible material is freely reusable.

Text and data mining introduces additional complexity. Legal exceptions and opt-outs vary by jurisdiction and purpose. Model training, retrieval indexing, and public generation may implicate different rights. The programme should record the provenance and permitted uses of source corpora. A private retrieval index does not automatically authorise redistribution of source content in generated chapters.

Licensing strategy should support preservation and interoperability. The canonical public text and metadata benefit from clear licences that allow libraries and independent platforms to preserve and render them. Proprietary lock-in undermines the claim that the book is a scientific record. Protected components can remain restricted, but the public structure, identifiers, and manifests should be portable.

Ownership inside collaborations needs early agreement. Partners may contribute background IP, jointly create results, or have different exploitation goals. The living environment can attach ownership and access metadata to artefacts as they are created. This reduces the final-stage scramble to determine who can publish what. It also makes generated derivative works easier to govern.

AI agents complicate ownership but do not remove human and institutional responsibility. A coding agent may generate a component influenced by training data or external code. A writing agent may produce passages close to a source. Similarity, licence, and provenance checks are therefore part of release validation. No automated detector can guarantee absence of infringement, but the system can reduce obvious risks and preserve evidence of due process.

The IP architecture serves the scientific vision by making protection explicit and time-aware. Without it, teams will either avoid living publication because it is too risky or release material carelessly. The goal is not maximum secrecy. It is a controlled path from private invention and collaborative work to citable public knowledge.

8.3 The public book as a governed boundary object

The public living research book is best understood as a boundary object between the programme and its audiences. It is detailed enough to support serious understanding, stable enough to cite, flexible enough to generate multiple interfaces, and selective enough to protect legitimate interests.

The phrase generated boundary object emphasises that the book is produced from a richer source under policy. Generation is not synonymous with autonomous writing. It includes selection, transformation, redaction, aggregation, explanation, and assembly. Each transformation should preserve links to the source and declare material loss.

A thousand-page edition can therefore be both extensive and incomplete. It may explain every prerequisite needed to understand the programme while omitting raw participant data. It may document all public algorithms while withholding a proprietary optimisation. It may describe the evidence for a security result while delaying exploit details. Completeness is evaluated relative to the claims and audience, not relative to the entire workspace.

The boundary object must resist two opposite failures. The first is privacy theatre: the programme invokes confidentiality broadly and publishes only conclusions. The second is transparency theatre: it releases enormous archives that no reviewer can interpret, then claims that everything was available. The living book aims for structured sufficiency.

A disclosure statement should accompany each major edition. It explains the classes of withheld material, governing reasons, review access, independent validation, and expected future release. It also identifies where restrictions materially limit reproducibility or novelty assessment. This statement becomes part of scientific evaluation.

The book can use graduated assurance labels. A claim may be supported entirely by public evidence, by controlled evidence reviewed independently, by proprietary evidence reviewed only internally, or by an attestation. The label does not replace explanation. It lets readers identify where trust is concentrated.

The programme should periodically test whether restrictions remain necessary. Patent filing may permit release. A privacy-preserving derivative may become available. A licence may be renegotiated. Security information may be safe after remediation. The living character of the book allows openness to expand over time.

Conversely, new risk may require limiting an interface. A public chatbot might reveal more than the static edition through inference. A newly discovered re-identification technique might make a dataset unsafe. The programme can update access while preserving the historical record of what was previously available. Such changes require transparency because they affect reproducibility.

The boundary object also mediates institutional trust. Reviewers, funders, regulators, industrial partners, participants, and the public do not receive identical views. The system should prevent silent divergence in factual claims across views. A confidential report may contain more detail, but the public summary should not contradict it. Cross-view consistency becomes a validation task.

The design is particularly relevant to public-private research. Companies may hesitate to expose workspaces, while academic partners need publishable outputs and career credit. A shared living book with controlled evidence layers can make collaboration more credible. It cannot resolve conflicts of interest, but it makes the publication boundary negotiable and auditable.

The strongest claim of this chapter is therefore normative. Scientific transparency should be judged by whether the published object allows appropriate scrutiny of its claims, not by whether every internal trace is open. Total exposure can be unethical and scientifically useless. Selective disclosure can be rigorous when the selection is principled, documented, and independently reviewable.

The boundary object remains credible only when its technical controls are embedded in governance that anticipates failure and abuse.

No architecture eliminates the possibility of concealment, leakage, or abuse. Policy labels can be wrong. Redaction can fail. Secure environments can be compromised. Reviewers can misuse access. Partners can dispute ownership. AI agents can infer sensitive information. Technical protection must therefore be accompanied by governance and remedies.

The first governance requirement is clear authority. The programme defines who can classify data, approve public release, grant reviewer access, decide IP timing, and respond to incidents. High-risk decisions require separation of duties. The person seeking publication should not be the sole authority determining that protected evidence is unnecessary for review.

The second requirement is audit. Access to controlled evidence is logged. Release decisions are recorded. Changes to classifications are versioned. AI systems used on protected material are identified. Audits should focus on meaningful risks rather than generating maximal logs that nobody examines.

The third requirement is challenge. Reviewers and readers need a way to contest a restriction. An editor or independent board can assess whether confidentiality is proportionate. Participants and data subjects need channels for concerns. Partners need dispute resolution. The publication cannot be accountable if the programme is the only judge of its own secrecy.

The fourth requirement is incident response. A privacy leak, premature disclosure, or licence violation triggers containment, notification, correction, and impact analysis. The living book can update affected interfaces and mark editions, but it cannot erase copies already distributed. Release design must assume that public disclosure is effectively irreversible.

The fifth requirement is preservation. Controlled evidence may need to remain available to qualified future reviewers even after the programme ends. This requires trusted repositories, access policies, and funding. A public claim that depends on a private drive scheduled for deletion is not durable science.

Technical claims should be modest. Encryption protects confidentiality under defined conditions; it does not ensure lawful purpose. Differential privacy bounds

one class of leakage; it does not make an unethical study acceptable. Confidential computing protects data in use against parts of the infrastructure; it does not protect against a malicious authorised analysis. Cryptographic hashes prove integrity relative to a value; they do not prove that the original data were valid.

The governance system must also recognise power. Large institutions may obtain privileged access unavailable to independent reviewers. Commercial partners may classify inconvenient information as secret. Participants may have little influence over secondary uses. A living publication architecture can reproduce these inequalities while appearing procedurally sophisticated.

The response is not a universal openness mandate, which would create other harms. It is to expose the distribution of access and authority. The book should state who could inspect what, under which conditions, and with which conflicts of interest. Independent public institutions, libraries, data trusts, and regulators can provide counterweights.

By 2040, privacy and IP will not be peripheral compliance sections added before release. They will be embedded in the scientific object. Every claim will carry an access topology: what evidence exists, where it resides, who has seen it, what can be reproduced publicly, and what remains protected. The credibility of the living research book will depend on making that topology legible.

9. CAREERS AFTER PUBLISH OR PERISH

9.1 Publish or perish, contribution portfolios, and programme-level evaluation

Publish or perish is commonly described as pressure to produce too much. The deeper problem is that it specifies what must be produced. When papers are the dominant countable unit of scientific value, researchers organise work into paper-shaped fragments. The metric does not merely observe science. It changes its architecture.

A long-lived programme has incentives to divide coherent work into multiple outputs, protect subtopics as individual territories, and delay integration until each contributor has secured a recognisable publication. Negative findings, maintenance, replication, data stewardship, and conceptual synthesis are difficult to convert into prestigious paper counts. Collaboration is praised rhetorically while career systems reward individually legible ownership.

This fragmentation has epistemic costs. Readers reconstruct programmes from scattered articles with overlapping introductions and inconsistent terminology. Results that depend on each other appear as independent contributions. Changes of mind are hidden because each paper presents a local finality. Software and datasets are cited less reliably than articles. The integrated theory may never be written because integration consumes time that could produce additional countable outputs.

Journal prestige amplifies the effect. The San Francisco Declaration on Research Assessment argues that journal-based metrics such as the Journal Impact Factor should not be used as proxies for the quality of individual articles or researchers (DORA, 2012). The Leiden Manifesto calls for quantitative evaluation to support, not replace, qualitative judgment and emphasises field differences,

transparency, and local context (Hicks et al., 2015). CoARA similarly calls for recognition of diverse outputs, practices, and activities, with peer review and qualitative judgment at the centre (CoARA, 2022).

These reforms identify the problem correctly, but implementation remains difficult because institutions lack rich, standardised evidence about contributions. A committee under time pressure falls back on journal names, citation counts, and paper totals. The living research book can supply a more inspectable object: not a narrative CV alone, but a verified portfolio of claims, experiments, tools, datasets, reviews, corrections, and maintained knowledge.

The book also exposes how artificial intelligence can intensify the current regime. If AI reduces the cost of literature review, drafting, coding, analysis, and submission, researchers can produce more papers. The local advantage is rational. Yet the collective result may be more competition, more review burden, more duplicated prose, and narrower exploration. The 2026 Nature analysis by Hao and colleagues reported precisely the possibility that individual advantage and collective contraction can coexist (Hao et al., 2026).

The response cannot be a moral demand that researchers publish less while institutions continue to reward publication volume. The unit of evaluation must change. A programme book offers a candidate because it preserves the integration that paper counts destroy and the granularity that collective authorship obscures.

This does not imply that every researcher should be evaluated by the number of pages contributed to a book. Page production would be an even worse metric in an AI-rich environment. Evaluation should focus on the significance, rigour, durability, and role of contributions within a research object.

A researcher who develops a method used across the programme has a visible contribution. A researcher who identifies a foundational error and leads the correction has a visible contribution. A researcher who maintains a data resource or reproducibility suite has a visible contribution. A researcher who produces an independent paper can still be recognised. The living book expands the evidence available to judgment rather than replacing one count with another.

A different evaluative object is therefore required: the contribution portfolio generated from the maintained programme but curated for qualitative judgment.

The future career dossier should be generated from contribution records but curated by the researcher. It presents a small number of significant contributions in context. Each contribution points to an edition, claim, artefact, review history, and role record. The researcher explains intellectual significance, independence, collaboration, and learning.

This portfolio aligns with the direction of responsible research assessment. It recognises diverse outputs while retaining qualitative review. A committee can inspect evidence rather than infer quality from venue prestige. External evaluators can review selected objects without reading an entire programme book.

Programme-level evaluation complements individual portfolios. A research organisation or funder assesses whether the programme produces coherent, reliable, reusable knowledge; maintains its infrastructure; corrects errors; shares appropriately; and develops people. The book becomes evidence of organisational capability, not only a publication.

The contribution portfolio can recognise work across time. A dataset maintained for ten years appears as a continuing contribution with releases, adoption, and quality history. A theoretical insight may mature through several editions. A negative result may later prove decisive. The system avoids treating each calendar period as an isolated list of papers.

Independence remains important, especially for early-career researchers. Granular records can show which claims or systems they led within a large collaboration. Letters and qualitative review remain necessary because significance and intellectual ownership cannot be inferred mechanically. The portfolio gives reviewers better material for those judgments.

Teaching and mentoring can connect to the book. A researcher who develops a tutorial, supervises a reproducibility project, or helps another team adopt a method contributes to knowledge transmission. These activities can be recorded without equating every educational interaction with a publication.

Peer review becomes visible scholarly work. Reviewers can receive credit for substantive critiques, replications, and edition assessments, subject to confidentiality choices. A review that prevents a false claim from entering the canonical book may be as valuable as a minor publication.

Maintenance becomes a career path rather than invisible service. Scientific software engineers, data stewards, research infrastructure specialists, knowledge curators, and epistemic editors can build portfolios of durable contributions. This supports team science by recognising roles that universities often place outside conventional academic progression.

The model also supports mobility. Contribution records use persistent identifiers and portable formats. A researcher leaving a programme retains a verified account of their work without taking protected data. The institution retains the objects and ongoing responsibilities. This separation reduces disputes over authorship and ownership.

Evaluation committees need new competence. Reviewing a contribution portfolio is more demanding than counting papers. Institutions must allocate time, train reviewers, manage conflicts, and resist pressure to collapse rich evidence into a score. Responsible evaluation is not cheaper by default. AI can assist navigation and consistency checks, but human judgment remains central.

The portfolio also needs proportionality. Junior researchers should not be required to maintain elaborate metadata for every activity. The programme environment captures evidence as work proceeds, and the researcher selects significant contributions. Administrative burden is a failure condition.

9.2 Gaming, surveillance, inequality, and collaboration

A richer evaluation system will be gamed. This is not an argument against it; every assessment system creates incentives. It is a reason to design against predictable failures.

The first failure is contribution inflation. Teams may create many small objects and attach roles to each. Agents can generate apparent activity at negligible cost. Evaluation should therefore ignore raw counts and focus on selected contributions, external use, evidential quality, and programme judgment.

The second failure is metadata capture. Senior researchers or institutions may control how contributions are recorded. Junior members may be under-credited. The system needs confirmation, dispute resolution, and independent oversight. A ledger is not fair merely because it is structured.

The third failure is surveillance. Fine-grained traces can reveal work patterns, communication frequency, failed attempts, and periods of low visible activity. Managers may use them to rank researchers. Scientific contribution records should be separated from operational telemetry. The default public and evaluative view contains meaningful approved contributions, not behavioural exhaust.

The fourth failure is conformity. If evaluation rewards contribution to large canonical books, researchers may avoid independent or disruptive work. A programme can become intellectually dominant and absorb credit. Career systems must preserve room for small teams, speculative projects, and work that challenges established research worlds.

The fifth failure is infrastructure inequality. Wealthy institutions can build polished living books with powerful agents, secure review systems, and professional editors. Smaller teams may appear less rigorous because their interface is poorer. Evaluation should distinguish scientific quality from production resources, and public infrastructure should reduce the gap.

The sixth failure is platform dependency. If a few commercial systems control contribution records and portfolio rendering, they can shape evaluation and extract sensitive data. Open standards, exportability, institutional repositories, and public governance are necessary. The scientific career record should not be locked inside one vendor.

The seventh failure is narrative capture. A beautifully integrated book can make a programme appear more coherent and important than it is. Committees must inspect the contribution delta and evidence, not reward literary completeness. AI-generated polish should be treated as interface quality, not scientific merit.

The eighth failure is automated conservatism. Evaluation agents trained on past successful careers may reproduce existing biases and prefer familiar contribution patterns. Their role should be limited to organising evidence and detecting inconsistency. Decisions about promise, originality, and significance require accountable human judgment and active bias review.

The transition will be uneven. Papers will remain important because databases, hiring systems, and disciplinary norms are built around them. Researchers who invest in living books may be penalised during the transition. Funders and

institutions that sponsor experiments must provide explicit recognition and bridge mechanisms.

A sensible path is dual accounting. Researchers continue to receive conventional citations and publications, while programmes pilot contribution portfolios and edition-level assessment. Comparative studies examine whether the new system improves fairness, collaboration, reproducibility, and intellectual diversity. The reform should be evaluated, not simply proclaimed.

The same infrastructure that creates new risks can also change the collaboration game once the paper no longer monopolises visible credit.

The strongest promise of the living research book is not better CV formatting. It is a change in the collaboration game. When teams can recognise contributions inside a shared object, they have less need to convert every subproject into a separate territorial paper.

A programme can organise around questions and capabilities rather than publication ownership. One group maintains a dataset, another develops methods, another performs validation, and another integrates theory. Their contributions remain visible in the common book. Papers are generated when a compact release serves the community, not solely to make work count.

This could reduce salami slicing and duplicate background. It could encourage researchers to improve shared explanations and infrastructure because the work remains attributable. It could make correction less threatening because identifying and fixing an error creates recognised value.

The effect is not automatic. Competition for high-status claims may persist. Contribution records can create new territorial boundaries. Leadership can still appropriate credit. The technical system must be accompanied by norms and governance that value collective outcomes.

The book also changes collaboration between generations. Senior researchers contribute synthesis, judgment, and historical understanding. Early-career researchers contribute new methods, experiments, and criticism. Technical specialists contribute infrastructure. The shared object makes interdependence visible without denying individual agency.

International and interdisciplinary collaboration may benefit because the book maintains shared definitions, access policies, and version history. Teams working under different legal regimes can expose compatible public views while retaining local protected data. Translation agents can support communication, with human review where concepts do not map cleanly.

The forecast for 2040 is not that publish or perish disappears as a phrase. Incentive systems change slowly. The forecast is that serious institutions will have an alternative object rich enough to make the phrase obsolete in practice. Researchers will still need to produce. What they produce can be durable knowledge inside a maintained programme rather than an endless sequence of fragments.

10. SCIENCE IN 2040: INSTITUTIONS, FAILURE MODES, AND THE TRANSITION

10.1 A concrete day in 2040 and its institutional ecology

It is 2040. A multi-institutional programme working on privacy-preserving clinical decision systems is preparing its biennial edition. The programme has existed for nine years and has survived two funding cycles, a major change in regulation, the departure of its first scientific director, and the rejection of one of its original hypotheses.

The researchers do not open a blank document. They open the programme environment. The current accepted claims, unresolved questions, evidence status, active experiments, data restrictions, software releases, and pending review issues are visible. The interface is not a single dashboard. It is a set of views over a maintained research object.

During the night, literature agents identified twelve potentially relevant publications. Three are duplicates or low quality. Five extend background sections without affecting the programme. Two challenge a benchmark assumption. One presents a similar architecture and may weaken a novelty claim. One reports a clinical subgroup failure that is directly relevant to deployment. Domain researchers review the proposed mappings. The novelty claim is narrowed. The subgroup result triggers a new validation task and a warning in the live book.

A coding agent has rerun the executable examples against updated dependencies. Most pass. One simulation differs from the previous edition because a numerical library changed. The agent isolates the cause, proposes a pinned environment, and marks the affected tutorial as unstable until a maintainer approves the correction. The public companion will not use that simulation in explanations during the unstable period.

A privacy agent reviews a proposed chapter update. The draft includes a generated case study based too closely on a rare patient trajectory. The agent flags re-identification risk and replaces the example with a validated synthetic composite. A human privacy reviewer approves the change and records why the synthetic example is pedagogical rather than evidential.

An IP agent notices that a technical optimisation described in a planned public video has not passed the invention gate. The video release is blocked. The optimisation remains in the controlled vault while the programme and its industrial partner decide whether to file a patent, retain a trade secret, or release openly. The book's public change log states that an implementation improvement exists but does not disclose enabling details.

The edition board meets. It does not read the whole book. It reviews the epistemic delta: four new claims, two weakened claims, one retired claim, three major corrections, a new dataset release, changes to the privacy analysis, and an unresolved disagreement about causal interpretation. The board assigns targeted external review.

Reviewers enter a secure cockpit. The public book provides a complete conceptual account, but one clinical claim depends on controlled data. The reviewers can run approved analyses inside a secure environment and inspect subgroup aggregates. They cannot export row-level data or upload the material to external models. Their AI assistant is hosted within the review boundary, records source paths, and is prohibited from making final recommendations.

One reviewer challenges the programme's interpretation. A sceptical agent independently finds the same weakness. The team runs an additional analysis, which does not resolve the issue. Rather than smoothing the disagreement, the new edition states it explicitly and lowers confidence. A future experiment is defined as the discriminating test.

The publication compiler produces several outputs. The canonical edition is 1,146 pages, including prerequisites, methods, evidence, limitations, and historical development. A thirty-page novelty report describes changes since the previous edition. A ten-page funder report maps progress to objectives and explains one abandoned milestone. A public audiovisual introduction is generated in six languages. A doctoral tutorial includes executable exercises on synthetic data. A

regulator interface exposes the compliance and evidence chain. A concise paper announces the main methodological result and cites the edition and claim identifiers.

No individual reads all 1,146 pages. A clinician asks for the evidence relevant to a particular subgroup. A machine-learning researcher explores the algorithmic contribution and re-runs the public benchmarks. A student follows a prerequisite path. A funder inspects objective changes. A historian compares the 2034 and 2040 editions to understand how the programme abandoned its original theory.

The book is not a decorative output. It is the programme's shared account, review surface, reporting substrate, onboarding environment, and public memory. Its existence changes how the programme works because important decisions must be connected to claims and evidence. Yet the private workspace remains larger. Internal speculation, protected data, detailed partner negotiations, and much operational history are not public. The edition is extensive, not exhaustive.

This scenario is technologically plausible by extrapolation from systems already visible in 2026: retrieval-grounded synthesis, research agents, executable notebooks, living reviews, research objects, secure data environments, continuous reporting, contribution taxonomies, and AI-assisted publishing. What does not yet exist is their reliable integration into a recognised scientific institution. The forecast is therefore primarily institutional.

This daily workflow is possible only because the institutions around the programme have changed their roles.

Journals will not simply disappear. Their most durable function is not paper distribution but certification, selection, editorial judgment, and community identity. In the living-book ecosystem, a journal may review a claim set, certify an edition delta, host adversarial evaluation, or curate a field-level synthesis. The journal article remains as a compact record of a scientific event, but it points into a larger maintained object.

Some journals may specialise in review services. They assemble expert panels, provide secure AI support, maintain reviewer reputation, and issue machine-readable assurance statements. Their value lies in trustworthy process rather than exclusive possession of text. Open peer-review journals may publish the review graph; confidential journals may preserve controlled attestations.

Publishers can become infrastructure providers for editorial integration, identity, rights management, accessibility, preservation, interface generation, and quality assurance. The danger is that they recreate enclosure at a higher level. A proprietary platform could control the chatbot, contribution records, and review environment even when the book text is nominally open. Portability and public standards are therefore conditions of legitimacy.

Libraries gain strategic importance. They preserve frozen editions, manifests, research objects, executable environments, and access policies. They may operate institutional companions grounded in archived editions and provide independent rendering when commercial interfaces disappear. Their role expands from preserving documents to preserving queryable scientific states.

Universities and research organisations maintain programme workspaces, contribution records, secure review infrastructure, and career portfolios. They need governance separating scientific coordination from personnel surveillance. They also need durable identifiers and export mechanisms so that researchers can carry verified contributions across institutions without carrying protected data.

Funders move from document-based monitoring to evidence-based programme review. They can inspect how objectives, claims, and risks evolve. They can require public editions and controlled evidence profiles proportional to the project. They can fund maintenance, independent review, and preservation as first-class activities rather than assuming that publication costs end with a paper.

Funders also become responsible for protecting exploratory research. Continuous visibility can create pressure to justify every deviation. Mature governance allows programmes to maintain private exploratory zones and report consolidated learning rather than live operational telemetry. Accountability should not eliminate scientific freedom.

Regulators and ethics bodies interact with relevant views of the book. A data-protection authority can inspect processing lineage and safeguards. A medical regulator can examine validation and change history. An ethics board can see whether protocol deviations affected participant risk. These bodies do not need the entire scientific narrative, but they benefit from the common source.

Professional societies define domain profiles. They specify what claims, evidence, provenance, and review status mean in their fields. A mathematical proof

object differs from a clinical trial, a qualitative interpretation, or an engineering benchmark. The living-book architecture succeeds only if it supports plural epistemologies while preserving enough common structure for citation and access.

Standards bodies maintain identifiers, packaging, provenance, contribution, access, and version semantics. Existing efforts such as FAIR, RO-Crate, persistent identifiers, CRediT, and responsible assessment principles provide foundations (Wilkinson et al., 2016; Research Object Community, 2025; CRediT, 2026; DORA, 2012). The future system should extend rather than replace them unnecessarily.

Libraries, public infrastructures, and research communities must also preserve plural interfaces. If one AI companion becomes the default interpreter of a field, it can shape understanding and suppress disagreement. Independent agents should be able to operate over the same public edition. The canonical source remains open to human inspection and alternative computation.

The institutional landscape is therefore not a replacement of publishers by AI. It is a redistribution of value from scarcity of text to stewardship of trust. Institutions that provide preservation, review, secure access, governance, and intelligible interfaces remain necessary. Those that merely control distribution lose justification.

Table 5. Institutional roles in the 2040 ecosystem.

Dimension	Implication for the living research book
Journals	Certify claim sets, edition deltas, adversarial reviews, and community significance rather than merely distribute text.
Publishers	Provide editorial integration, secure review, rights management, accessibility, provenance, and interface quality under portable standards.
Libraries	Preserve frozen editions, manifests, executable components, access conditions, and independent reading interfaces.
Universities	Operate programme environments, contribution records, secure data services, and qualitative career assessment.
Funders	Review evidence and objective evolution, pay for maintenance and review, and limit access to contractually justified views.
Professional societies	Define domain-specific evidence, review, and publication profiles.

Dimension	Implication for the living research book
Regulators and ethics bodies	Inspect authorised compliance and evidence views derived from the same canonical source.

10.2 Failure modes of the living research book

The first failure is the cathedral of fluent emptiness. AI makes it easy to construct an imposing object with extensive background, elegant transitions, generated diagrams, and apparent comprehensiveness. The core contribution may be trivial. Length and polish create an authority effect that overwhelms reviewers and public readers.

The defence is architectural. Every edition begins with a contribution map and an epistemic delta. Claims have explicit novelty and evidence status. Background is labelled as synthesis rather than contribution. Reviewers can evaluate the centre before entering the cathedral. No metric rewards page count.

The second failure is the coherent false world. AI is exceptionally useful at resolving local inconsistencies in prose, but scientific reality may be inconsistent with the programme's theory. A system can integrate weak evidence into a persuasive narrative, smooth disagreement, and repeatedly cite internally compatible sources. Coherence becomes insulation.

The defence is adversarial plurality. Contradictory evidence, alternative interpretations, and rejected claims remain visible. Independent retrieval and external review challenge the programme's source selection. Claim graphs distinguish internal support from independent replication. Coherence is treated as a property to test, not a proof of truth.

The third failure is scientific monoculture. Retrieval systems repeatedly surface the most visible literature. Agents imitate established methods. Career portfolios reward contributions to dominant programmes. The efficiency of shared books can reduce exploratory diversity. The 2026 evidence that AI use can coincide with narrower collective attention makes this more than a theoretical concern (Hao et al., 2026; Traberger and van der Schaar, 2026).

The defence includes explicit exploration budgets, independent forks, diversity-aware literature search, support for small programmes, and evaluation that

values intellectual risk. A book should record where its literature coverage is weak and which perspectives are excluded. No agent can guarantee diversity, but the infrastructure can make concentration measurable.

The fourth failure is selective deception. Privacy and IP restrictions are used to conceal evidence that would weaken claims. Independent reviewers receive only curated outputs. Attestations become ceremonial. The public book appears rigorous while essential components remain inaccessible.

The defence is disclosure topology and independent governance. The book states which claims depend on restricted evidence and what assurance exists. Reviewers can challenge access decisions. Funders and editors can require stronger controlled verification. Persistent restrictions reduce claim status rather than disappearing from view.

The fifth failure is surveillance. Continuous contribution records, agent logs, and project intelligence become tools for measuring workers. Researchers alter behaviour to satisfy dashboards. Informal thought and dissent disappear. The system improves managerial visibility while damaging science.

The defence is data separation, minimisation, governance, and labour rights. Scientific provenance records meaningful contributions and decisions, not every action. Operational telemetry has limited retention and access. Individual evaluation uses curated portfolios. Researchers participate in defining policies and can challenge misuse.

The sixth failure is platform capture. A small number of companies host workspaces, models, review systems, and citation interfaces. They can change prices, terms, ranking, or training practices. A field's knowledge becomes technically open but operationally dependent.

The defence is open standards, exportable research objects, institutional hosting, model plurality, and library preservation. Public money should not produce scientific records that cannot be independently rendered or queried.

The seventh failure is maintenance collapse. The grant ends, the agents stop, dependencies break, and the living book becomes a frozen website. A thousand-page object without stewardship can decay faster than a PDF.

The defence is explicit retirement and preservation. Not every component remains executable forever. Frozen editions, manifests, source archives, emulation strategies, and migration plans preserve what can be preserved. Maintenance obligations and funding are assigned before programme closure.

The eighth failure is review overload at a higher scale. A larger object generates more possible checks and more issues. AI flags thousands of concerns. Human reviewers become responsible for an environment they cannot understand.

The defence is scoped assurance. Review focuses on material changes and high-impact dependencies. Automated checks are evaluated and prioritised. Programmes use profiles proportionate to risk. The system reduces burden only when it is designed to suppress low-value work.

The ninth failure is metric migration. Institutions stop counting papers and start counting claims, contributions, reviews, updates, or reuse events. Gaming returns in a new vocabulary.

The defence is to resist automatic scoring. Structured evidence supports qualitative judgment. Selected contributions matter more than totals. Evaluation periodically examines its own behavioural effects.

The tenth failure is epistemic centralisation inside teams. The editorial board controls the canonical narrative and marginalises dissent. The book's integration becomes a political technology.

The defence is versioned disagreement, forkability, contributor appeals, external review, and visible governance. A canonical edition can state a current position without erasing alternatives.

These safeguards do not guarantee success. They make the proposal falsifiable. If living books increase volume without improving understanding, if they centralise power, weaken privacy, or reduce diversity, the institution should be revised or rejected. A future worth constructing must remain open to evidence against itself.

Table 6. Principal failure modes and design responses.

Dimension	Implication for the living research book
Cathedral of fluent emptiness	Expose the contribution delta, claim status, and evidence independently of length and polish.

Dimension	Implication for the living research book
Coherent false world	Preserve contradictory evidence, branches, external review, and independent retrieval.
Scientific monoculture	Measure source concentration, fund exploratory branches, and maintain plural interfaces and model families.
Selective deception	Publish the disclosure topology, permit challenges to restrictions, and lower assurance when evidence remains inaccessible.
Surveillance	Separate scholarly contribution records from behavioural telemetry and impose retention, access, and labour protections.
Platform capture	Require open standards, exportability, institutional hosting, and library preservation.
Maintenance collapse	Define release, retirement, archival, migration, and funding responsibilities.
Metric migration	Use structured evidence to support qualitative judgment rather than creating new automatic counts.

10.3 Transition, evaluation, and the wager

The transition will not begin with thousand-page books. It begins by changing how projects preserve and connect what they already produce.

Between 2026 and 2029, research teams can pilot structured programme records. Claims, evidence, code, data, literature, and contributions receive stable links. Existing standards such as FAIR, RO-Crate, CRediT, DOIs, ORCID, software citation, and data citation are used where possible. AI agents assist curation and literature monitoring. The public output remains mostly papers and reports.

The first experimental books should be generated for programmes that already produce many related publications and deliverables. They can integrate prerequisites, results, methods, and limitations into a coherent edition while pointing to existing repositories. Reviewers compare whether the book reduces time spent reconstructing context. Teams measure maintenance burden and error detection.

Between 2029 and 2033, controlled review capsules and publication compilers mature. Programmes produce annual editions alongside conventional papers. Funders accept generated reports from the maintained object. Libraries begin

preserving manifests and machine-actionable editions. Contribution portfolios are piloted in hiring and promotion.

Research is needed on retrieval fidelity, claim extraction, provenance, privacy leakage, redaction quality, IP detection, secure review, version semantics, contribution fairness, and human use. The programme should publish negative results about the infrastructure itself. It should compare AI-assisted living books with well-maintained conventional documentation rather than with neglected repositories.

Between 2033 and 2036, journals and societies develop certification profiles for programme editions and claim releases. Review becomes more modular and continuous. Public companions become common, but independent interfaces and source export remain requirements. Some disciplines adopt living books faster than others. Computational and data-intensive fields may lead; fields dependent on tacit craft or sensitive human contexts will require different designs.

Between 2036 and 2040, the model becomes plausible as a default for major research programmes. Papers persist as release notes, priority records, and community signals. The living book becomes the primary explanatory and evaluative object. Annual or biennial editions support funding, careers, teaching, transfer, and public communication.

This timeline is a construction programme, not a forecast of inevitable market adoption. Institutional resistance may be stronger than technical difficulty. Paper counts are convenient. Journals confer status. Researchers understand existing career rules. Funders prefer templates. Publishers protect business models. Transition requires actors who benefit from the new object and are willing to bear early costs.

The strongest early adopters may be research programmes that already suffer from fragmentation: large consortia, interdisciplinary infrastructures, long-lived software and data projects, regulated research, and programmes with extensive public reporting. Their need for integration is immediate.

Small projects should not be forced into maximal infrastructure. A lightweight living book may contain a few chapters and a compact research object. The principle scales: granular source, coherent account, stable editions, selective disclosure, and traceable interfaces. Complexity should follow scientific need.

Evaluation of the transition needs clear criteria. Does the system reduce reviewer time without reducing review quality? Does it improve reproducibility and correction propagation? Does it preserve knowledge when staff change? Does it recognise contributions more fairly? Does it reduce redundant reporting? Does it protect privacy and IP better than current ad hoc practice? Does it increase or decrease intellectual diversity? Does it create new inequalities?

The tools built now should encode the possibility of failure. They should export data, expose provenance, support alternative models, and allow projects to retire components. A research programme committed to living books should not become captive to its first implementation.

The transition culminates in a wager specific enough to guide present engineering and specific enough to be proven wrong.

The wager of this book can now be stated without excessive caution.

By 2040, the article will no longer be the primary intellectual home of serious, long-lived research programmes. It will remain useful as a compact publication event, but the canonical object will increasingly be a living research book: extensive, versioned, AI-maintained, multimodal, executable where appropriate, and connected to a structured research substrate.

These books will often be hundreds or thousands of pages because source completeness and reader burden will have been decoupled. People will read generated routes rather than entire volumes. Reviewers will inspect contribution maps and evidence capsules rather than perform literature archaeology. Funders will query maintained knowledge rather than reconcile duplicated reports. Teams will receive credit through contribution records rather than author order alone. Libraries will preserve editions and machine-actionable structures. Publishers and journals will earn legitimacy through certification, review, preservation, and secure interfaces rather than control of distribution.

The public book will not expose the entire workspace. It will be an epistemically sufficient publication compiled under privacy, confidentiality, security, licensing, and IP policies. Reviewers may receive controlled access beyond the public edition, but access will be scoped. Scientific transparency will mean that important claims can be evaluated under explicit conditions, not that every internal trace is public.

The future will not arrive cleanly. AI may instead accelerate the production of conventional papers until the literature becomes even more difficult to trust and navigate. Living books may become corporate knowledge portals without public accountability. Contribution systems may become surveillance. Monographic integration may become a machine for smoothing disagreement. These outcomes are plausible.

The reason to articulate the positive alternative is that technological systems are built around prior assumptions. If we assume that the paper remains the final unit, AI will optimise paper production, paper review, and paper metrics. If we assume that a research programme can maintain a richer public object, we will build provenance, selective disclosure, claim-level review, version governance, and contribution infrastructure.

Foresight is useful when it changes present design. The claim of this monograph is not that we possess a crystal ball. It is that the current scientific record is fragmented for historical reasons that AI can weaken, and that a coherent alternative can be constructed from technologies and reforms already emerging.

Reality may contradict the forecast. That possibility is not a weakness. It is the condition that makes the forecast scientific enough to guide research. The programme should build prototypes, compare outcomes, expose failures, and revise the institution. A future publication system that cannot be corrected would betray the very principle of the living book.

The final proposition is therefore both prediction and invitation. The future of scientific publishing may contain much more text while requiring much less reading. It may become more granular for machines and more integrated for humans. It may protect private workspaces while giving reviewers better evidence. It may replace the paper's monopoly without discarding the paper's virtues. The living research book is the form through which these apparent contradictions can become one scientific institution.

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